



UNHCR Environment



**SETTING UP AND DEVELOPMENT
OF AN ENVIRONMENTAL
DATABASE FOR REFUGEE
PROGRAMMES**

PHASE 1 : AFRICA

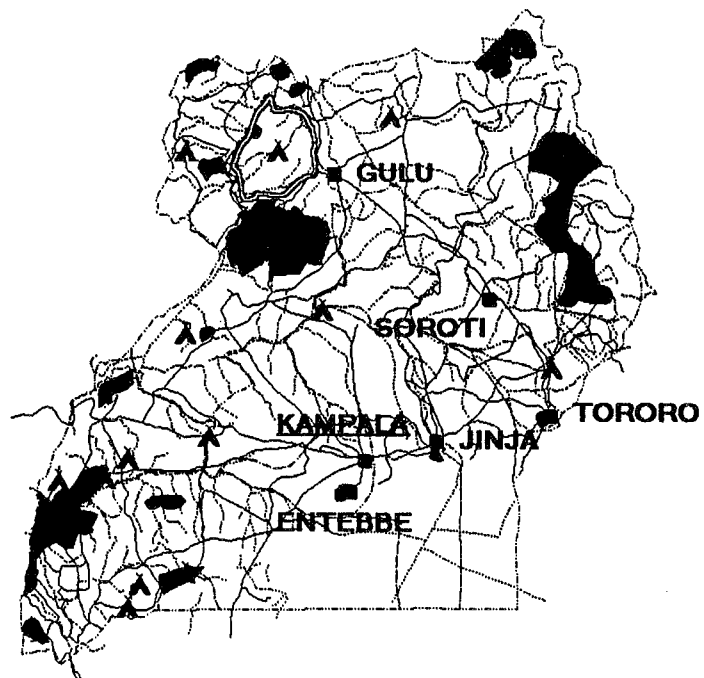
**First results
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**UNHCR's REFUGEE SITES, PROTECTED AREAS and
WILDERNESS AREAS in UGANDA**



ABSTRACT

An environmental database concerning most of the UNHCR's refugee sites across the world was established using the Geographical Information System (GIS) technology.

This database aims to provide UNHCR, within the development of its environmental policy, an "environmental identity card (in terms of sensitivity)" for each refugee site. A better systematic knowledge of ecological conditions for each site will follow from this information.

From this analysis comes an environmental cartography, which should facilitate UNHCR's actions in this domain, notably concerning prevention and planning. The first stage of this analysis concerns Africa.

The first results essentially touch upon the living conditions of the refugees (type of climate) as well as the potential impacts of the refugee populations on certain components (establishment of refugee sites within or close to wilderness or protected natural areas).

This first phase of the project leads us to propose, in conclusion, a certain number of recommendations for the conditions concerning the development of the database, our perspective being an operational environmental policy within the UNHCR.

The opinions expressed in this report are those of the author and not necessarily of the UNHCR itself.

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LIST OF ABBREVIATIONS

CRSC:	Canadian Remote Sensing Center
FAO:	Food and Agriculture Organization of the United Nations
GIS:	Geographical Information System
GPS:	Global Positioning System
GRASS:	Geographic Resources Analysis Support System (USA)
GRID:	Global Resources Information Database of UNEP
IGN:	Institut Geographique National (France)
IUCN:	International Union for Conservation of Nature and Natural Resources
LGP:	Length of Growing Period
NASA:	National Aeronautical and Space Administration (USA)
NOAA:	National Oceanic and Atmospheric Administration (USA)
PTSS:	Program and Technical Support Services, UNHCR Geneva
UNDP:	United Nations Development Programme
UNEP:	United Nations Environmental Programme
UNHCR:	United Nations High Commissioner for Refugees
USGS:	United States Geological Survey
WCMC:	World Conservation Monitoring Center
WHO:	World Health Organization
WRI:	World Resources Institute

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I. INTRODUCTION

1.1 General Context

After the Rio Earth Summit (held from 3-14 June 1992) and Agenda 21 which was a result of this conference, the international community has become aware that within the framework of lasting development for current, but especially future generations, it is vital to preserve what falls under the general term: the environment. In the spirit of this conference, this generic term includes such issues as: biodiversity, desertification, protection and management of eco-systems etc.

Action aimed at protecting the environment must be taken in all areas. It must be taken within macro-economics activity as well as within specific target areas.

Among all the factors known to disrupt ecological equilibrium (over-exploitation of natural resources, pollution, etc.), there still remains one relatively unrecognized by the general public and the international community, but which can have (or has already) important, indeed dramatic, consequences on the environment and more particularly on natural resources.

This vital factor concerns populations falling under the term of refugees and displaced persons. Their number (refugees: 18.4 million and persons displaced within their own country: 24 million) increased from 1986 to 1990 from 26.4 million to 42.4 million (Dennis, 1993; Westing, 1992; UNHCR statistics, 12/31/92).

The problem is aggravated by the fact that these populations are often found in already fragile ecological areas (in a developing country) where environmental problems do not necessarily count among the priorities for development.

The United Nations High Commission for Refugees (UNHCR) was created in 1951 as a subsidiary organ of the UN General Assembly. Its mandate is "to assume the function of international protection...to refugees...and of seeking permanent solutions for the problems of refugees." A refugee being a person who "is outside the country of his nationality...because he has a well-founded fear of persecution by reason of his race, religion, nationality or political opinion and is unable or because of such fear, is unwilling to avail himself of the protection of the government of the country of his nationality...and includes also persons fleeing from civil war (i.e. actual conflict in Rwanda or in the former Yugoslavia)."

The flow of people which can affect thousands, hundreds of thousands, indeed millions of people--as is currently the case in Rwanda--does not take place without environmental consequences.

Indeed, a high population concentrated in a limited space inevitably leads to environmental problems.

Among the most often cited and most crucial environmental problems concerning refugees, we have:

- problems related to deforestation;
- the destruction of grass lands and animal habitats;
- problems related to health, water and sanitation.

From another perspective, the environmental conditions themselves can have harmful consequences for the refugees. The concentration of refugees within a hostile environment (a barren, dry climate for example) will exacerbate the population's own problems as well as those of the host environment.

The UNHCR's policy concerning the environment is still not clearly defined. As underlined by S. Gurman: "UNHCR currently deals with environmental issues on an 'ad hoc' basis; it has no policy or guidelines and does not employ consistent environmental planning, monitoring or evaluation instruments."

In the face of these shortcomings, UNHCR has now begun the process of defining a policy on refugees and the environment which takes into account the unique conditions under which it operates (S. Gurman, Y. Stevens, J. Dennis).

In 1992, during the 43rd Session of its Executive Committee, UNHCR highlighted the basic principles of its environmental policy.

"In view of the potential environmental hazards related to refugee assistance programs and the need to sustain the environment in refugee hosting areas, the whole spectrum of the Office's activities...must become more sensitive to environmental considerations for refugees and local populations for durable solutions...the office will seek to:

- plan and implement refugee assistance programs in an environmental way;
 - adopt an ecologically sensitive approach to refugee assistance programs;
 - play a catalytic role in encouraging other agencies and institutions to address large-scale environmental problems in refugee-hosting and returnee areas."
- (Y.Stevens, 1993).

On this basis, an Environment Co-ordinator has been seconded by the Japanese government to elaborate this policy and co-ordinate efforts to address environmental projects. The Interim Guidelines for Environment-Sensitive Management of Refugee Programmes was recently issued, which marked the first step for such policy development.

1.2 Objectives

One of the main projects of the Co-ordinator is the development of an environmental Database (at a scale of 1: 1M) for refugee sites under the care of UNHCR.

Following the recommendations of the previous study (M. Stauffacher, 1994) and as laid out in the Terms of the Reference, the objectives of this consultancy are:

- 1 to collect the basic environmental information (i.e. elevation, temperature, precipitation, vegetation, type of soil, desertification risk, etc.) on all refugee sites identified and georeferenced during the first phase;
- 2 to set up a tabular database for the information collected as above and organize for selected countries, a country-specific cartographic package for environmental and other planning purposes;
- 3 to design a statistical analysis scheme in terms of vulnerability for the refugee sites on a global scale.

1.3 Methodology

To carry out these objectives, we present the different stages which lead us to accomplish these tasks. The report will take the following course:

- I The tools used: GIS MapInfo for Windows and the computer configuration;
- II Entering and establishing the database, presenting the data, the kind of data, the integration of the data into MapInfo and the constraints of this integration;
- III Organizing the data within the database and establishing a country-by-country Database;

- IV Proposal for a qualitative environmental approach, taking into consideration selected parameters, to produce maps useful for decisions in planning and prevention; and
- V List of recommendations for the growth and development of the database.

Observations and preliminary notes:

Establishing an environmental policy within UNHCR is a burdensome and difficult task.

Just beyond the budgetary problems, which often remain the major issue, two large problems present themselves when discussing the environment and the management of natural resources.

- First, in urgent and pressing situations, environmental problems are not prioritized;
- Second, refugee situations are by nature unpredictable and planning is short-term by design, whereas the management of natural resources must be considered over the long term (CURE 1993).

The advantage of possessing a Geographic Information System (GIS), within the framework of an UNHCR environmental policy, is vital (Dennis, 1993).

The system must allow those in charge at the UNHCR:

- 1 to quickly access geographic and environmental information, on a given scale (local, national, regional, global) for all the countries of the world;
- 2 to have a synthesized global vision --in the form of maps and related tables-- of the geographic and environmental conditions of documented refugee sites;
- 3 to quickly obtain synthesized maps, useful for making decisions (according to the necessary scale of the data).

The advantages are the following:

- 1 To take stock, in a systematic, renewable and operational manner, of all the refugee camps in the world and the corresponding environmental conditions around them (desertification, type of climate, etc.).
- 2 In emergency situations, to quickly provide a topographical and environmental evaluation of the place considered (cf maps of Rwanda).

- 3 Within the framework of an environmental policy, one can better approach solving problems and discovering causes and then set up plans and strategies, thus promoting environmental projects whose aim is to find lasting solutions for refugees, local populations and the surrounding environment.

From the standpoint of an environmental policy, GIS should help to provide answers in the three following fields of research:

- 1 **Localization and Verification**
(i.e. how many refugee sites are concerned)
- 2 **Appraisal**
(the degree and kind of damage)
- 3 **Prevention**
(scenarios and solutions)

II. INTRODUCING THE GIS MapInfo

GIS is a database and decision-making tool.

As underlined by M. Stauffacher in his report, when interested in geographic analysis, the use of GIS is a modern, powerful and efficient tool.

It is one of the vital components in setting up an environmental policy within UNHCR (Dennis 1993).

We will not go over the technical aspects of GIS (cf Stauffacher 1994); however, it is important to make those non-initiated aware of the fact that the GIS is not just a black box which outputs "pretty maps" but is a real, decision-making tool, which can be used in different fields of investigation and research.

It is above all at the core of a geographic and environmental data bank which must be developed on a global scale as well as on a local scale (multi-scale approach, J. Dennis, R. Blake, J. Kamangira), as part of a real environmental policy within UNHCR.

In this report, GIS is not only a cartographic software package but a new separate component, in the means possessed by the HCR for its actions.

Numerous agencies--the UN among others--already have a service tied to GIS within the scope of their actions: UNEP, FAO, WHO, UNDP, IUCN, WRI, World Bank (UNEP/GRID report, 1992).

With the expanding development of computer tools and their performance (speed, storage capacity, user-friendliness, etc.), the proliferation of GIS tools also tends to increase rapidly. Parallel to this expansion, the development of numerical data bases, notably cartographic, increases considerably. Thus numerous international organizations (UN), national and/or private agencies are now systematically developing databases of different scales (UNEP/GRID, NASA, Canadian Center of Remote Sensing, EEC, AGRHYMET (Niger), DRSRS (Kenya), etc.).

Moreover, these databases are more and more easily accessible, notably via the Ethernet network (UNEP/GRID, WCMC, NASA, ...). With this technique, one can thus easily examine them and, according to access rights, repatriate the selected data in real time.

2.1 MapInfo

MapInfo is a software program which belongs to the family of vector-type software (cf M. Stauffacher 1994, Burrough 1983).

It allows rapid viewing of precise geographic information, linear or surface, for a given area.

Relatively "simple" to use, it allows easy access to synthesized information regrouped in the form of a spreadsheet, which is called an attribute table and is linked to geographic components.

The information is accessed:

- either under the form of topological requests, (one examines the database on the level of its graphic elements i.e.: selection of sites situated within a 50 km radius from a given city);
- or by attribute requests, (one directly examines the table, in a more or less complex way depending on the type of request. These requests are made with the help of a particular language called SQL (Structured Query Language), i.e.: location of all the sites in Kenya which possess a population greater than 10,000 refugees).

2.1.1 Advantages/Limitations of the software:

Advantages:

- MapInfo is a relatively inexpensive software package considering its capacities;
- it is relatively easy to access (in its base operations, especially for map display) for people uninitiated in GIS;

- it allows transfer of information from other databases: Excel, Lotus, and/or dBASE, and it links them directly to graphic components;
- it possesses an important cartographic projection module (more than 300 projections) which largely facilitates the location of information (use of maps employing varied projections to locate sites);
- it allows for digitalization, that is, direct integration of new geographic information using a number table.

Limitations of the software:

- Integration of graphic data (and associated attributes) is difficult without the downloading module ArcLink;
- it does not allow all of the operations relative to the "real" GIS (ArcInfo, IDRISI,...), notably, it does not allow (for the time being) balancing of maps with the objective of obtaining a resulting map according to a mathematical model (i.e. erosion map of soils via the model (i.e. erosion map of soils via the model USLE using soil maps, slopes and land uses);
- it does not allow, in the current version, an installation into the network at a reasonable cost.

III. SETTING-UP THE DATABASE

3.1 Type of Data

3.1.1 Data relative to the refugee sites

In the study undertaken by M. Stauffacher, information relative to the refugee sites was collected. This information comes essentially from activity reports of UNHCR, published by continent and from different interviews between the consultant and the Desk Officers.

The information, **the original reference being the refugee sites**, which is integrated into the database concerns: the continent, the country, the capital, the name of the refugee sites, the number of refugees (if available), the date of establishment of the sites (if available), the latitude and longitude of the sites (information essentially obtained either from a geographic atlas or from the database ArcInfo/INFO of the UNEP/GRID Geneva), and commentary relative to the types of sites particular to the typology of the UNHCR.

In order to have the most pertinent information possible, we have added the date, which concerns the information communicated on the sites, to this table. Therefore the updating is done using this as a base.

As the geographic information concerning the position of the sites is often difficult to obtain, a certain number of sites where this information is missing (for example, there is no site location map for Angola) are not integrated into the database.

In this phase of the project, the database is not exhaustive concerning all of UNHCR's refugee sites in the world.

As mentioned by M. Stauffacher, "until now, elementary information about sites has never been centralized elsewhere, so this database must be the start of such a process, which is very important for future projects."

Remarks concerning the geographic position of the refugee sites :

There exists no database in UNHCR relative to the geographic position of refugee sites throughout the world. This is a serious deficiency which the project has brought to light and which it aims to remedy.

For this reason, the research of geographic information remains difficult and approximate.

This situation leads to :

- a lack of coherence in the geographic location of refugee sites: we don't know exactly how to locate the refugee sites;
- logistical difficulties for UNHCR because the geographic position is uncertain;
- within GIS, erroneous environmental information when one examines the database, (i.e. a geographic localization of a site which varies by 100 km in a piedmont region leads to false information, notably for temperatures and precipitation - notion of gradients).

It is advisable to make an effort to have exact knowledge about the refugee sites; this concerns basic information essential for any UNHCR activity (inventory, planning, and logistics...cf recommendations).

3.1.2 Data pertaining to "regions" comprised of a refugee population or of displaced persons

By using UNHCR reference sources (UNHCR report 1992/1993), we also integrated (whenever it was possible, if maps existed) data pertaining to regions with refugee populations or displaced persons, but not taken into account by UNHCR. The notes

concerning the exact geographic location of these zones are identical to those on locating the refugee sites.

3.1.3 Data from UNEP/GRID

For this first phase, the main source of digital maps came from UNEP/GRID Nairobi. For this first stage, data concerning Africa was ordered for various reasons: technical (incompatible data format for some data - raster data), or independent of our wishes (delays in Nairobi). A certain amount of this data still has not reached UNHCR in Geneva.

At present, we have the following data:

Administrative Units for Africa from UNEP/FAO Database (no date)

- scale: 1:5M
- type of information: country, second and third level administrative units, area, perimeter
- format: vector from ArcInfo coverage
- type of feature: polygons
- geographic projection: latitude/longitude

Main roads for Africa from UNEP/FAO Database (no date)

- scale: 1:5M
- type of information: types of roads: motorable/paved roads, length of lines
- format: vector from ArcInfo
- type of feature: lines
- geographic projection: latitude/longitude

Africa Surface Hydrography from UNEP/FAO Database (no date)

- scale: 1:5M
- type of information: length of segment; feature coverage: OC: stream segment through ocean, L :Stream segment across large land mass, stream situation : 0: stream segment through waterbody, 1: Stream segment across land, (note: the file is incomplete)
- format: vector from ArcInfo
- type of feature: lines
- geographic projection: latitude/longitude

Railroads for Africa from World Data Bank II (US State Dept., no date)

- scale: 1:5M
- type of information: length of segment
- format: vector from ArcInfo
- type of feature: lines
- geographic projection: latitude/longitude

Protected Areas for Africa from World Conservation Monitoring Center (WCMC), International Union for Conservation of Nature and Natural Resources (IUCN) (1987)

- scale: 1:1M
- type of information: protected areas* by country, effectiveness of the protection in the area (good, moderate, poor, no protection)
- format: vector from ArcInfo
- type of feature: polygons
- geographic projection: latitude/longitude

Wilderness areas of Africa from World Bank and Sierra Club (no date)

- scale: 1:2M
- type of information: African wilderness areas**, area, perimeter
- format: vector from ArcInfo
- type of feature: polygons
- geographic projection: latitude/longitude

Population Density for Africa (1988)

- scale: 1:5M
- type of information: area, perimeter, population density
- format: vector from ArcInfo
- type of feature: polygons
- geographic projection: latitude/longitude

Agroecological zones for Africa from UNEP/FAO Database (no date)

- scale: 1:5M
- type of information: country, type of climate, Length of Growing Period (LGP), population and population development (by agro-eco zones), density of population (by agro-eco zones), animal units, density of animal unit by hectare, area, perimeter
- format: vector from ArcInfo
- type of feature: polygons
- geographic projection: latitude/longitude

* Protected areas : parks and reserves of Africa (from IUCN classification).

** Wilderness areas : areas showing no evidence of development (i.e. settlements, roads, or other human works), from World Bank classification.

3.1.4 New parameters that we are awaiting for Africa

Socio-Economic parameters from American Digital Cartography (ADC is a private US company)

- scale: 1:1M
- type of information: 17 layers i.e. cities and villages, lakes and small lakes, powerlines and utilities, airports, elevation, cultural landmarks, hydrology, etc.
- format: ArcInfo converted into MapInfo
- type of feature: point, lines, polygons
- geographic projection: latitude, longitude

Environmental data from WCMC

- scale: 1:1M
- type of information: types of Forest and classified forests, types of wetlands all around the world
- format: vector from ArcInfo coverage
- type of features: polygons
- geographic projections: latitude/longitude

African population densities by administrative units from the University of California, Santa Barbara (UCSB)

Medium resolution database for the African continent, derived from UNEP similar dataset (World Atlas of Desertification, 1992). Estimated population densities by administrative units (second and third level, with estimated growing rates).

Other types of parameters can be added to the database from:

- other UNHCR departments (i.e. PCBS: statistical data about refugees' nationalities, PTSS for Land mines, etc.);
- other UN agencies (i.e. types of diseases from WHO, etc.);
- other organizations or agencies: World Bank, GTZ, etc.;
- other NGO's;
- national databases from all around the world (i.e. economic components, etc.).

3.1.5 Quality of the data

The generally acceptable margin of error for digitalization (according to the IDRISI manual) is 1/10 of a millimeter. For maps and charts on a scale of 1:1M and 1:5M, this means about 100 meters and 500 meters respectively.

Since the errors concerning locating the sites are greater, these margins can be considered acceptable.

3.2 Data Integration

3.2.1 Downloading Data

3.2.1.1 Downloading Graphic Data

The data coming from the majority of the surveyed organizations are in the ArcInfo format (cartographic and tabular data), the most widely used software package of its kind in the world (UNEP/GRID report, 1992).

This data cannot be directly downloaded into MapInfo. Thus, it is advisable to limit transformations which are too large from the ArcInfo files in order to reduce possible errors connected with reiterations.

In this case, only two ways exist for downloading the data:

- 1 change the ArcInfo files within ArcInfo into the DXF format (Drawing Interchange File), which requires having ArcInfo!;
- 2 possess MapInfo's ArcLink module, which allows the direct downloading of ArcInfo files in only one format: ".E00" (ArcInfo export file with no compression).

For this study, we chose the first solution. We wrote a program in collaboration with UNEP/GRID (Dr. B. Zand, D. Delpietro) in order to change the ArcInfo files into a format directly readable by MapInfo. In the meantime, ArcLink was acquired and will be directly used when the next ArcInfo files are received.

3.2.2 The constraints connected with downloading graphic data

Downloading files is not easy with the DXF format.

DXF is a format produced by graphic software packages (DAC type) and does not lend itself readily to the integration of attribute data as they exist in the majority of the GIS. This limitation leads notably to difficulties in recognizing certain types of objects.

If there is no special or particular problem regarding points and lines, then on the other hand there is a certain difficulty downloading polygons via DXF.

The DXF format "destroys" the structure of the polygon in order to transform it into a polyline (the polyline has the same shape of a polygon but it is not closed, that is, the arrival point does not have the same exact coordinates as the departure point). For this reason, the program written by the GRID consisted in automatically closing all of the polylines in order to transform them into a "region" (name of polygons in MapInfo) and conserved the relationships with the associated attribute tables.

3.2.3 Non-graphic data

MapInfo also allows for the downloading and the manipulation of data under the form of spreadsheets, output from different existing databases: Excel, dBASE, Lotus, etc. which have no associated cartographic component.

For example, mapping refugee sites (generated under EXCEL) resulted in a manipulation which consisted of the creation of a geographic map of refugee sites from information initially contained in a spreadsheet (using longitude and latitude of the sites).

In order to associate spreadsheet data to the maps, MapInfo offers two possibilities.

- 1 Geocoding: this operation consists in linking the table "without a map" to a file which has a table and an associated map, using a parameter which can be a column in common (for example, the name of the country).
- 2 "Create Points": this function allows integration of new pinpoint data (for example, the position of refugee sites), if the geographic coordinates (longitude and latitude) are known (map, GPS etc.).

The integration is done in decimal degrees (it is therefore necessary to convert degrees, minutes and seconds of the points) and according to the cartographic projection chosen (Lambert, UTM etc.).

Note: It is also possible to integrate the data via a digitalization program installed under MapInfo. This procedure (the most commonly used to integrate cartographic documents into a digital format) allows one to save, after having selected the desired projection, all types of cartographic information by digitising the elements (points, lines, polygons).

The cartographic information can then be linked to the attribute tables using a common range (for example, the identifier of the digitized element).

IV. ORGANIZATION OF THE DATABASE

The data provided by UNEP/GRID Nairobi are global data and concern the entire African continent.

As is specified in the recommendations (M. Stauffacher 1994), it is advisable to organize the databank by country. This idea seems logical as UNHCR organizes its work by geographic region by grouping together several countries (Desk Officers).

This organization of data allows one to provide a "country package" which is directly of use to the Desk Officers and all professionals. As outlined by J. Dennis, "GIS System must be implemented to provide information about projects and/or environmental data, at any time in an easy way for everybody."

In this perspective, GIS's work consists of extracting the specific information linked to each country from the general files (roads, rivers, agro-ecological zones, etc.).

In order to accomplish these operations, it is necessary to recall the specifics of the GIS. At this time, two methods can be used in order to select the necessary information for each country.

These two methods are:

- 1 either the existence, in the attribute data (the table) of a column specifying the country to which the geographic information belongs (in this case this is done using a SQL command (Structured Query Language); or
- 2 the use of geographic operators which act directly on the relationships maintaining the graphic objects between them (points, lines, polygons).

The first method is easier and is also exhaustive because it supposes that all information relative to a country is already classified using a column which specifies belonging to a certain country.

But in the files provided by the UNEP/FAO, the parameters (except for the file: administrative limits) are not classified by country. In this case, we must call upon the geographic operators.

The use of geographic operators must be done carefully, given that the operator works on the objects themselves. At the point of intersection (for example, the extraction of lines: the routes in the interior of the polygon: the country), it is possible that the last lines will not be recorded - especially those situated on the border of the polygon.

This occurs because the operator works on the centroide of the object. If the centroide is on the exterior of the polygon, this object will not be selected (it is the same for polygons). It is then advisable to verify the result with the initial file, and if the occasion should arise, to make another selection (Intersect) which acts (element by element) on the intersection between the objects.

With each operation (SQL or geographic) having to do with polygons, the file relative to the refugee sites is updated and added with the information received.

It is easy to examine the table relative to the refugee sites and to obtain instantaneously all of the relative environmental information that is associated with it (i.e., temperature, precipitation, length of growing period, etc.).

The information for the African continent was integrated by using one or both of these two operators.

The operation is illustrated by the following figure (GIS) :

- 1 SQL selection of chosen countries
- 2 Extraction file by file (roads, agro-ecozones, cities, etc.) of information relative to the countries selected
- 3 Verification of the "quality of the data"
- 4 Organization, in a directory, of all the information country by country

Figures 1, 2, and 3 illustrate these steps.

The updating of information (which we plan to do about every year - except for emergency situations or special requests) based on refugee sites will be done for the country concerned:

- either by elimination (and storage in archives) of obsolete information; or
- by integration, by country, of the new sites into a pre-existing file.

This second stage is done in the following manner :

- 1 by creating, momentarily, a new file containing new information (latitude and longitude etc.);
- 2 by incrementing this file with associated environmental information;
- 3 by deactivating - absence of geocoding - the initial file of the sites;
- 4 by adding the new information to the reference table (refugee sites) of the country concerned;
- 5 by geocoding the reference table which is updated by the new information.

Information concerning all 169 refugee sites on the African continent was coded, this consisting of systematically linking all of the environmental information to the refugee sites.

GIS Principles

type of operation

SQL Selection
from tables

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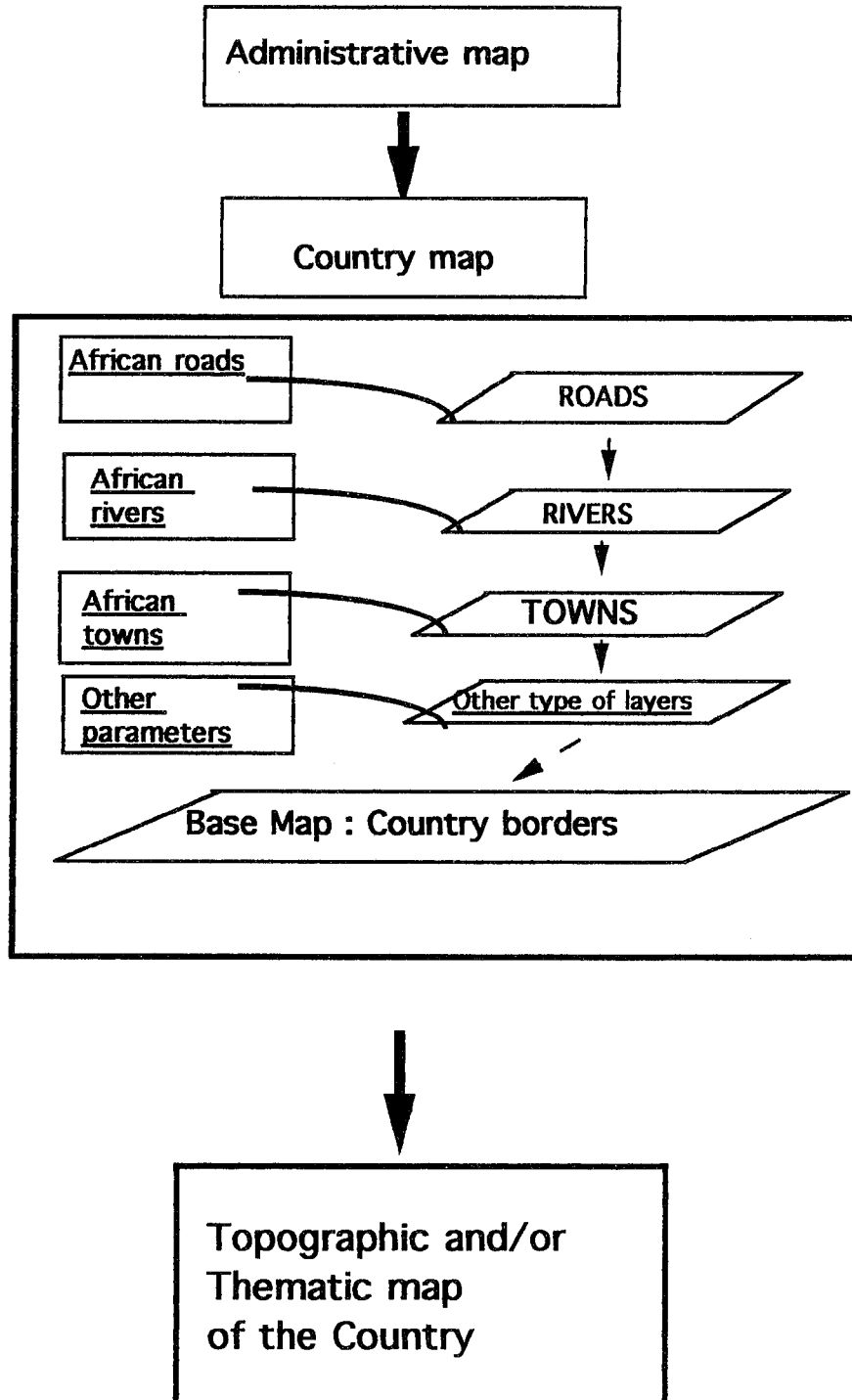
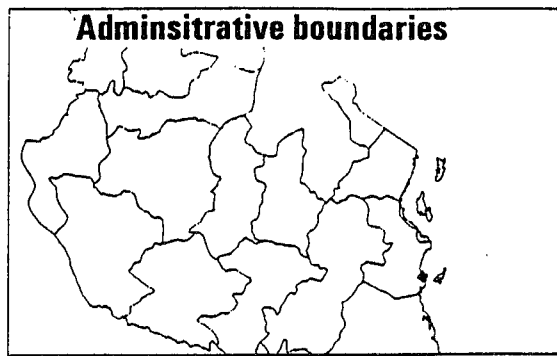
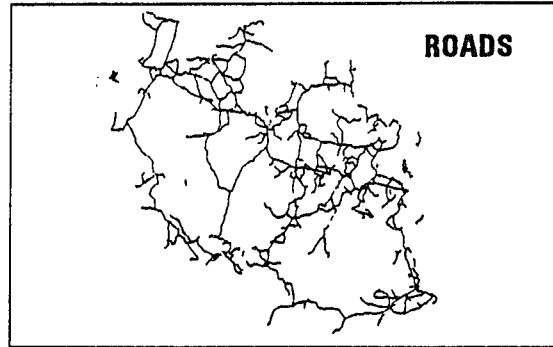


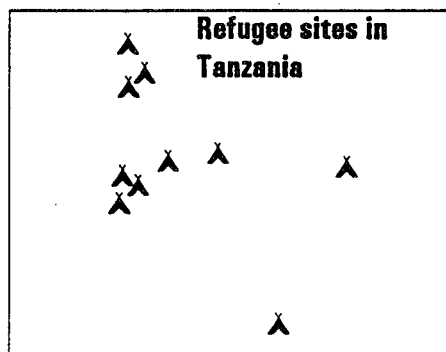
Fig 1



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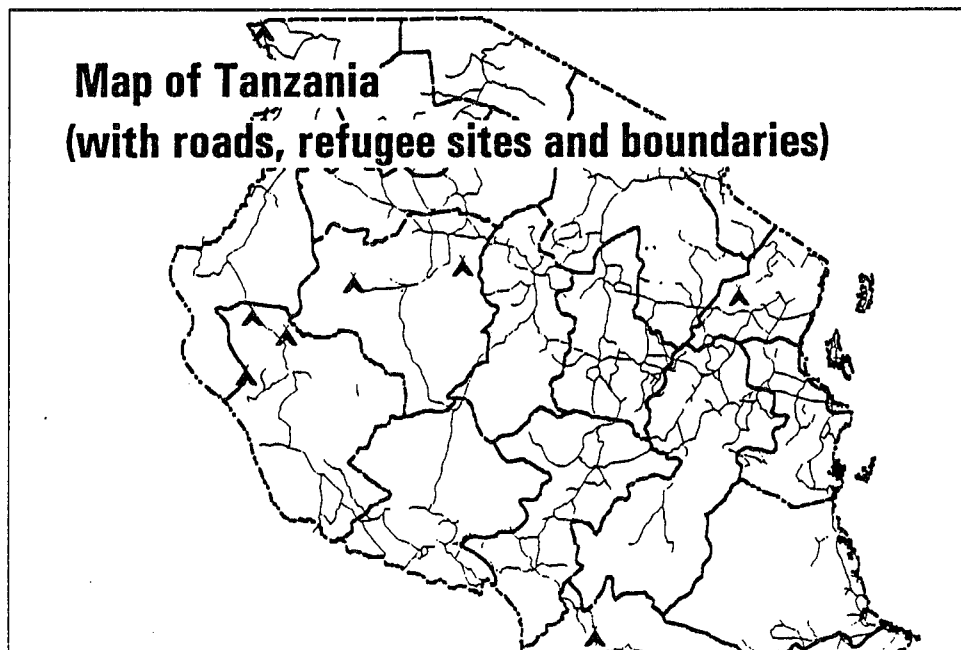


Fig 2

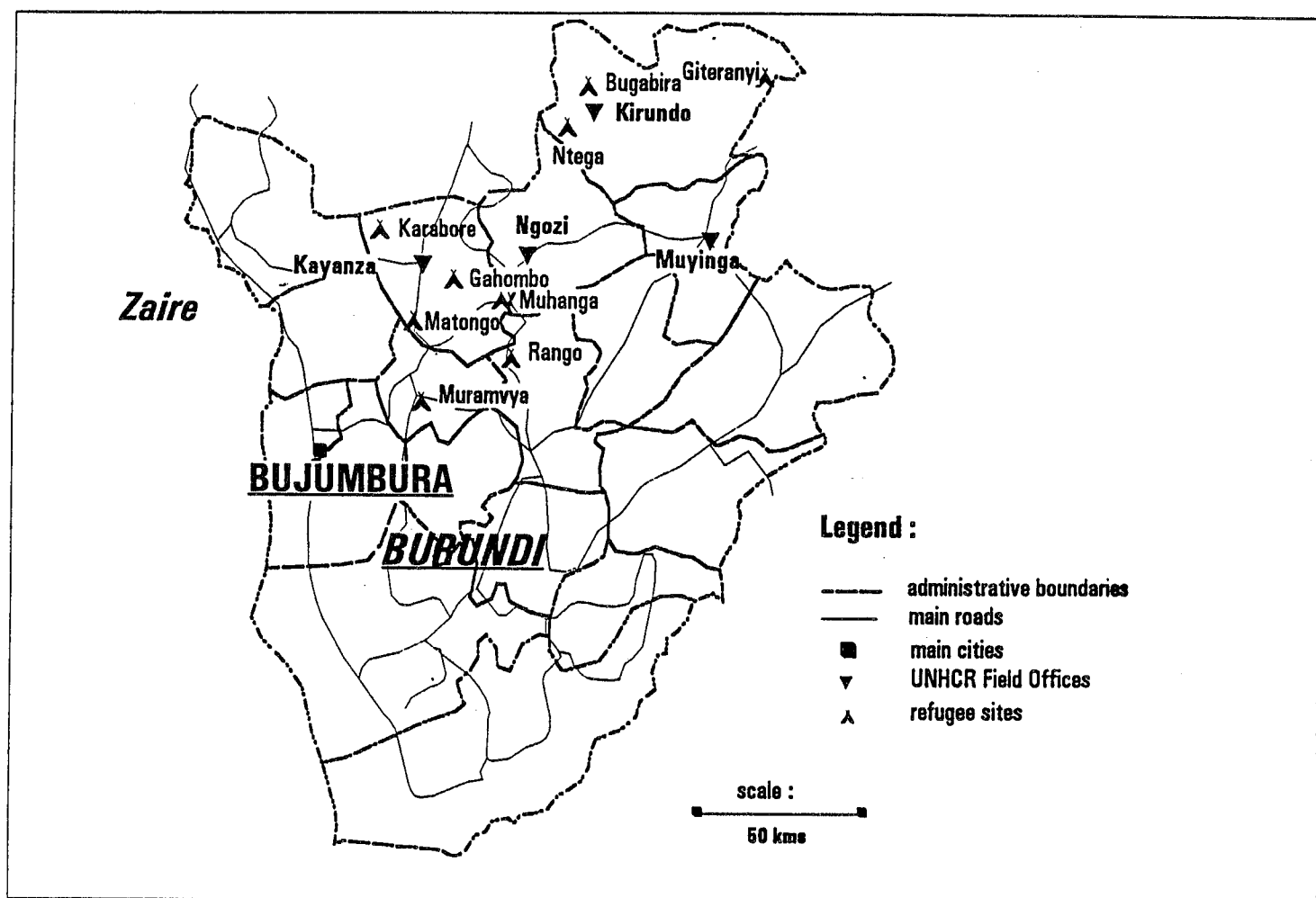


Fig 3

V. PROPOSAL OF A GLOBAL-SCALE STATISTICAL ANALYSIS SCHEME FOR THE ENVIRONMENTAL VULNERABILITY OF THE REFUGEE SITES, EXAMPLES OF APPLICATIONS

One of the most interesting applications of the GIS is without a doubt the decision and the cartography which follow from it.

The advantage of the database, as aforementioned, concerns :

- not only the notion of compiling pertinent information and examining the database according to more or less complex requests;
- but also, and above all, the production of a computer-assisted thematic cartography which will help all of the professionals of UNHCR in the following areas:
 - **topographic cartography:** location of sites, logistical analyses;
 - **thematic cartography:** selection of one or several parameters and then outputting maps showing which sites correspond to these criteria (sites located in a sensitive climate zone);
 - **decision-aiding cartography:** procurement of information, at the country level, on the favorable and unfavorable zones for the establishment of a refugee site.

The relationship between the refugees and their environment can be of several different natures. As outlined in various articles and reports^{***}, three types of relationships can be distinguished:

- environmental parameters which lead to difficult living conditions for the refugees;
- environmental impact linked to the establishment of a refugee population;
- migration of a population because of difficult environmental conditions.

^{***} S. Gurman, Y. Steven, J. Dennis, I. Muinul, V. Laissailly-Jacob, M. Zmolek.

5.1 The environmental parameters which generate difficult living conditions for the refugees

These can be regrouped under three essential themes :

- * **Climate parameters:** these have an important influence on the lives of the refugees. They can have a direct influence, such as precipitation, temperature, wind, etc. (It is essential to know the monthly distribution of these parameters or an indirect one such as the length of the growing period.)
- * **Pedological parameters:** types of soil will have a direct influence on agriculture and by consequence, on the lives of the refugee populations. Latrine building, foundations for buildings, roads and drainage are also affected. The combination of pedological and climate parameters can lead to other unfavorable conditions such as soil erosion.
- * **Demographic, socio-economic and ethnic parameters:** An already existing local population (human and/or animal) can generate harmful consequences for the refugee population. The availability of land for agriculture may be at stake; this can also cause tensions between the locals and refugees.
- * **Topography/relief:** Increased risk erosion
- * **Proximity to flood prone areas** (e.g. wadi rivers) (Flood hazard)
- * **Endemic diseases**

5.2 Refugees' impact on the environment

The arrival of a large population on a site can be the cause of serious consequences for the environment.

The refugee population can either be the sole cause of these consequences, or exacerbate the already existing local problems. (cf 5.1)

These consequences are essentially linked directly to the number of refugees concerned and the duration of the sites.

Possible causes of degradation:

- * **Soil and vegetation over-exploitation** which leads to soil degradation (i.e. loss of ground cover), erosion and desertification.

- * **Proximity of sites to protected areas, wilderness areas and forests** can lead to disturbance of these sites (i.e. deforestation, which in turn may lead to soil loss or the appearance of erosion spots (Mazowe Camp in Zimbabwe****). This soil loss disrupts water's capacity to infiltrate into the soil and reduces the ground-water resources (i.e. Malawi: CURE report 1993).
- * **Proximity of sites to reservoirs or small lakes** can lead to water pollution and disease (i.e. cholera in Zaire Goma region).

In addition to the parameters discussed in sections 5.1 and 5.2, two other important elements which disturb the well-being of the refugee populations are:

- * **emergency situations linked to refugees** can generate problems concerning living conditions, keeping in mind "the unpredictable emergency of the situation which often does not allow adequate time for environmental assessments of the impact of the population influx on the site",
- * **the lack of choice** for the selection of sites and settlements." (Y. Stevens)

5.2.1 Indirect implications linked to environmental problems

As outlined by Y. Stevens, indirect implications linked to these environmental problems also exist, notably :

- * **economic problems** linked to the scarcity of certain natural resources such as wood used for heat or cooking (scarcity here being tied to deforestation), or linked to poor soil quality which is in turn a cause of weak agricultural production. The need to go on long-distance searches for necessities, such as wood used for cooking, upsets the social and family structure of these populations as well.
- * **deforestation**, for example in Sudan (Wad Sherifée and Safawa sites), leads to erosion problems bordering the river banks, silting up of rivers (flood hazard). Deforestation can also lead to **wind erosion** (sites in the Ogaden region in Ethiopia) and the augmentation of dust storms leading to respiratory problems for the refugees.

**** K. Watanabe: Mission Report to Kenya and Zimbabwe (May 94).

note : as underlined by R. Black: "the arrival of large numbers of refugees may also have positive economic benefits, providing markets and/or labour to increase production in labour scarce areas, or leading to the flow of international aid to particular regions."

5.3 The environment as a cause of population migration

Environmental "events", for example, drought, the eruption of a volcano, floods etc., can cause internal migration (within a country) or international migration (like the Sahel in the 70's) (World Bank, Refuge).

5.4 Proposal of a qualitative methodological approach for the evaluation of camps located in environmentally sensitive situations

5.4.1 Identification

In order to establish an environmental policy, it is first advisable to identify all of the refugee sites located in environmentally or ecologically "sensitive"***** areas.

Because of the database that has been created, as well as the functions of GIS, we can quickly provide (for Africa at this time) an estimation of the refugees sites corresponding to these criteria.

There is not, strictly speaking, a simple general model of evaluation which would permit rapid estimation of the environmental context for a particular site. The existing models call for:

- either the particular parameters and index notions such as "refugee absorption capacities" developed by the WRI, or site resiliency capacities. These models deal only with one environmental component at a time and call for deduced information (especially on a large scale) which is not always easy to obtain (i.e. fertilizer use per capita, annual population growth rate etc.); or
- more general and more complex models, such as Environmental Assessment, which requires a certain quantity of information - extra-environmental or on a large scale. These models call for an in-depth expertise, which must be carried out at a local and a national level.

***** sensitive : i.e. corresponding to the criteria in sections 5.1, 5.2, and 5.2.1.

In the current state of the study, the methodology which we can apply in order to identify refugee sites, is essentially based on empirical and qualitative evaluation, using the parameters included in the database.

Thus, we present a preliminary evaluation concerning the refugee sites located in regions which, from an environmental point of view, can be characterized as "sensitive".

This evaluation must then be analyzed case by case as each refugee site has specific characteristics, should allow us :

- 1 to locate all of the sites which may be classed as "sensitive" according to the essential parameters described in sections 5.1, 5.2, 5.2.1;
- 2 to accomplish a rapid environmental estimation of selected sites according to the reports and studies already completed, in order to evaluate whether the selection corresponds or not to a "sensitive" situation. (pertinence of the selection);
- 3 according to bibliographic research, to interview the field officers in order to evaluate rapidly - in situ - according to selection criteria: LGP, temperatures, precipitation etc.. (feedback on the selected sites);
- 4 according to these responses and this documentation, the establishment of a preliminary typology of "sensitive" sites based on these selection criteria and the rough outline of a hierarchical order of intervention; and
- 5 following this typology, the elaboration of an environmental intervention policy.

The broad outline of this approach is organized in the following figure (fig. n° 4).

This flow-chart shows the rough outline of an evaluation approach of the sensitivities of a given site.

In the absence of all required environmental information (average temperatures, precipitation, etc.), it is difficult to propose a precise typology, especially concerning the choice of parametric classes to be considered. For the moment, however, we can nevertheless establish "an environmental identity card" of all the refugee sites and then propose a typology of sensitivities according to the already integrated parameters.

The following table indicates, in an empirical manner, (a more exhaustive bibliographic analysis must be undertaken to establish a pertinent selection of classes to be considered) the approach which we are able to consider according to existing parameters.

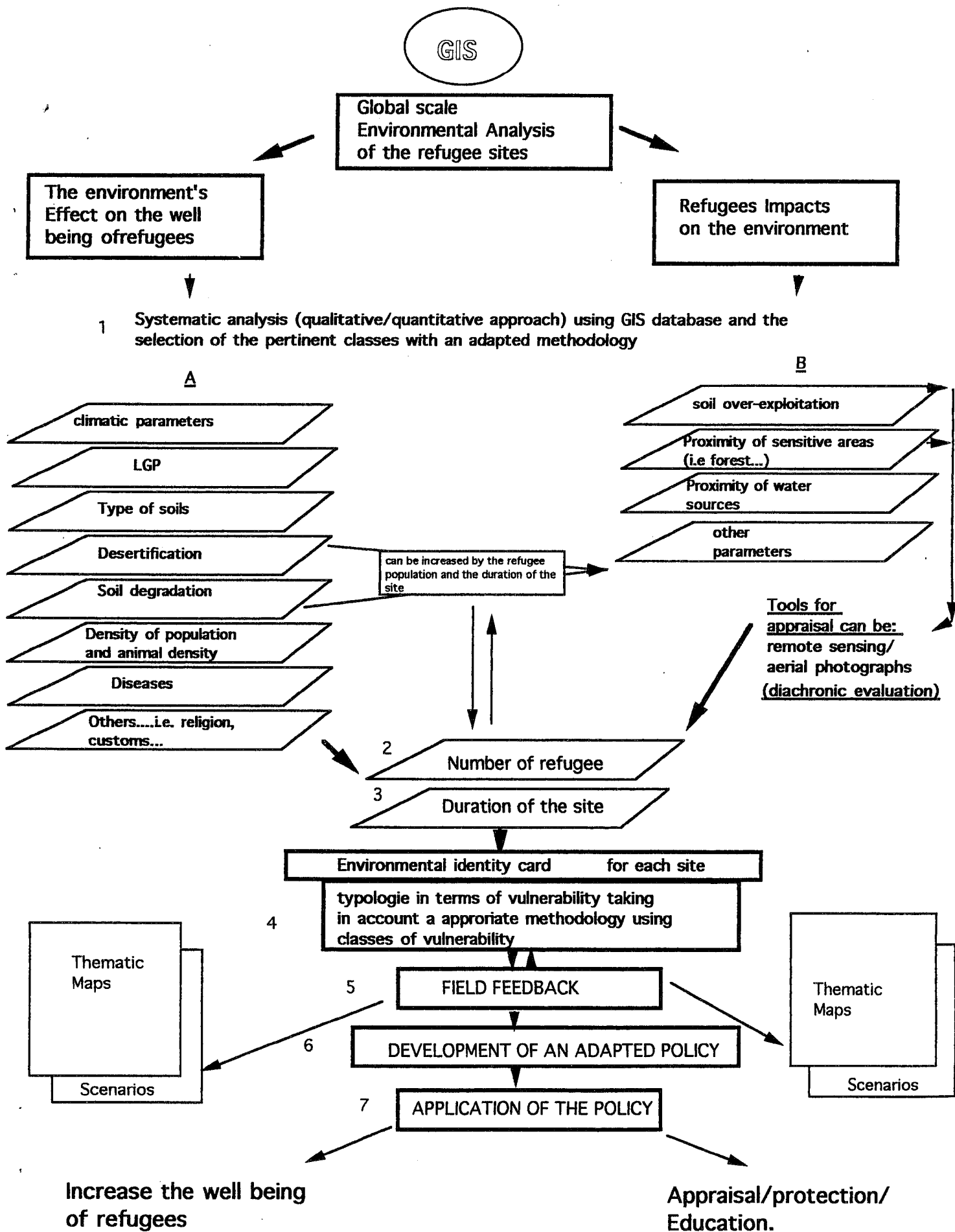


Fig 4

Note: The relative classes: for the LGP, the population densities, the proximity to wilderness or protected zones are given as an example, and in no way are the result of a particular modelization.

Parameters	LGP	sensibility	Density of population inh/km	sensibility	protected/wilderness areas	sensibility	number of refugee	sensibility	duration of the camp	sensibility
classes	0 days	very high	> to 500	very high	inside/border	very high	X relative	very high	X relative	very high
	1-75 days	high	>250-500	high	1-5 km	high		high		high
	> to 75	medium	<100 - 250<	medium	> to 5 km	medium		medium		medium

Table 1: Example demonstrating the steps to adopt according to the environmental parameters for the selection of the most "sensitive" sites

The selections are then made automatically within the GIS, with the help of SQL requests. The most sensitive sites are those which show the highest index: "very high" for one or more of the chosen parameters.

Regarding the number of refugees and the duration of the site, it's only after the first selection which bears on the environmental criteria (fixed factors) that we will choose and then will establish (after an - in situ - study of each site) a second typology which takes into consideration these two factors (evolving factors).

In order to illustrate this approach, we are presenting several examples of applications, accomplished using preliminary information compiled in the database.

Example 1: Evaluation of established sites in a problematic climate context.

For Africa, the database currently groups together 169 refugee sites.

Keeping in mind the information sources obtained and described in Chapter 2, a preliminary "environmental" evaluation was done on all of the African sites.

Many refugee populations are located in difficult environmental contexts, especially in regards to climate. (S. Gurman).

The interest in the database lies in the fact that it is able to instantaneously provide, according to pre-defined criteria, the list of sites corresponding to these criteria.

Having no monthly data (for the time being) relative to precipitation and/or temperatures, we have used information from the FAO file: agro-ecological zones, and more particularly, the criteria: length of growing period (LGP).

The length of growing period is defined as the time period in days over which the total rainfall plus an assumed soil moisture storage of 100 mm (P100) is more than half the potential evapotranspiration (PET): $LGP = P100 > 0.5 PET$ (J.H. Venema 1991).

The "LGP" is a very important agricultural parameter, as it defines the length of the season which is favourable to agricultural production. Thus, it is an essential piece of information for designing any agricultural projects for refugees. Within the context of the environmental assessment, this information is issued to determine the regeneration potential of vegetation and the viability of re-afforestation activities.

LGP might also be useful in determining viability of re-afforestation activities.

It is possible to make a correlation between the "LGP" and types of arid zones (UNEP/UNEP-GRID/CRU/UEA).

We have therefore defined 4 types of "LGP" which correspond to the 3 types of more or less arid zones.

LGP	Type of climatic zone	Number of sites concerned
LGP = 0 days	hyper-arid zone	13
0 < LGP < 75 days	arid zone	13
75 ≤ LGP < 119 days	sub-arid zone	37

Table 2: Number of refugee sites selected according to the type of climate

Figure 5 shows the Djibouti example. In this country, there are four refugee sites of more than 5000 people who live in difficult climatic conditions. The LGP = 0 days.

Figure 6 lists all of the selected sites in which the "LGP" is 0 days and < 75 days.

In this example, all other conditions being equal, the sensitivity level of sites is evaluated essentially by taking into consideration: 1. type of climate, 2. the number of refugees concerned, 3. the duration of the camps.

Among all of the selected sites, the Tinduff site in South Algeria seems to be in the most harsh condition as far as the living conditions of the refugees are concerned (reports PTSS 19/88, 93/13).

Obviously, with the available methodology, it is advisable to verify the information with precision, for example: nature of the soils, dimensions of the sites, the exact number of refugees, etc...

The refugees' impact on the environment

Example 2: Applications to protected zones (IUCN) and to "wilderness" zones (World Bank).

A refugee population affects the environment in many ways. We have partly cited these in the preceding chapter (5.2).

Refugee sites located in a hyper-arid zone (LGP = 0 days)

Sources UNEP/FAO

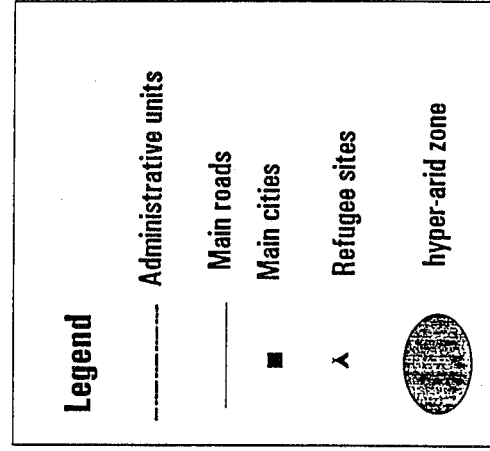
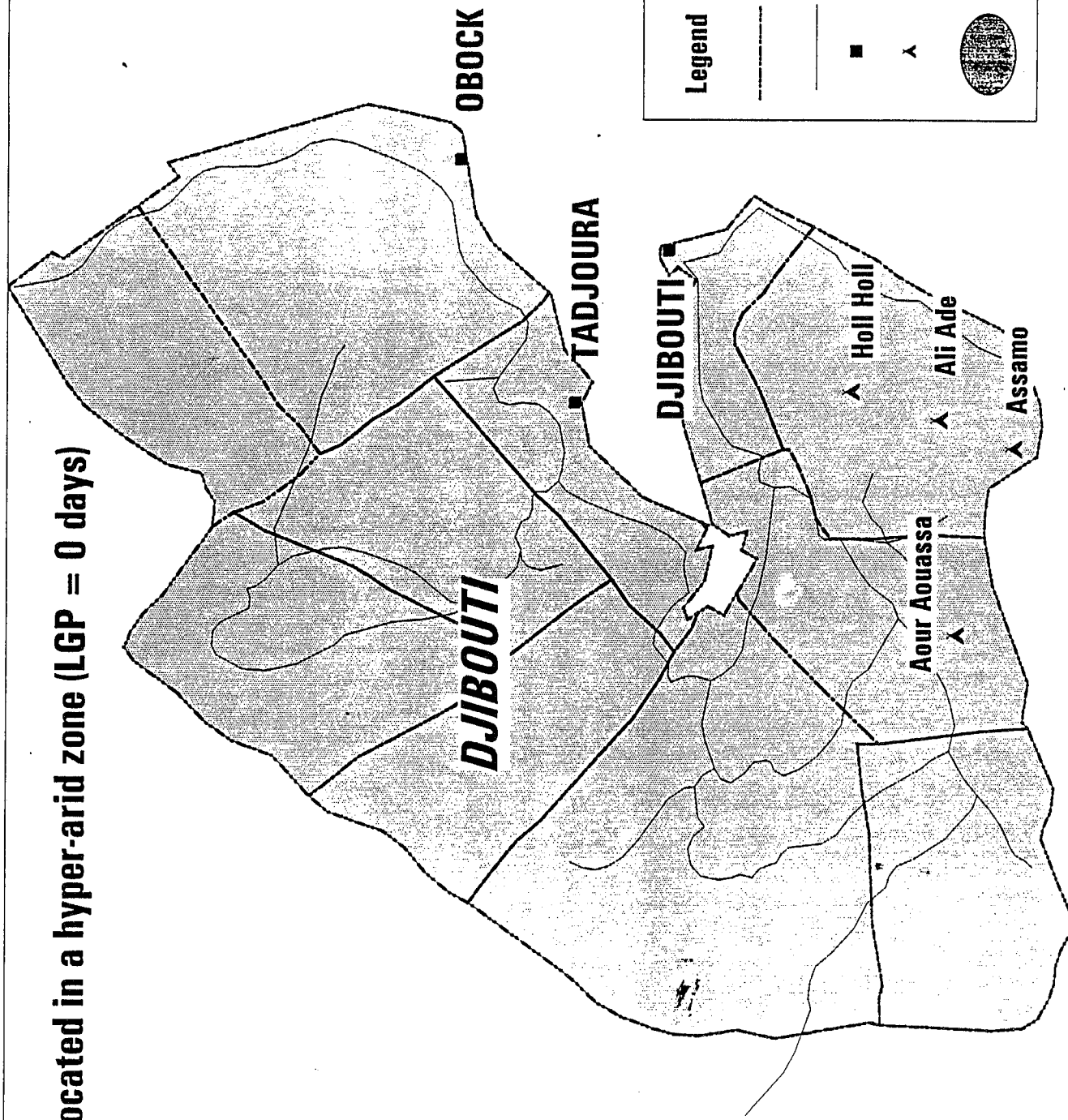


Fig 5

Continent	country	Capital	camps	refugee	year	Poptimesyear	longitude	latitude
Africa	Algeria	Algiers	Tinduff	80000	1975	1520000	-8.04	27.51
Africa	Djibouti	Djibouti	Assame	5300	0	0	42.82	11.01
Africa	Djibouti	Djibouti	Hol Hol	7200	0	0	42.92	11.29
Africa	Djibouti	Djibouti	Ali Ade	9900	0	0	42.87	11.14
Africa	Djibouti	Djibouti	Aour Aouassa	7800	0	0	42.49	11.11
Africa	Ethiopia	Addis Ababa	Daror	15000	1990	80000	48.6	7.71
Africa	Ethiopia	Addis Ababa	Kamaboker	25000	1990	100000	44.42	8.14
Africa	Ethiopia	Addis Ababa	Rabasso	10000	1990	40000	45.77	7.99
Africa	Ethiopia	Addis Ababa	Aisha	20000	1990	80000	42.37	10.56
Africa	Kenya	Nairobi	El Wak	8200	1990	32800	40.75	2.59
Africa	Kenya	Nairobi	Hagadera	43829	1990	175318	40.5	0.188
Africa	Kenya	Nairobi	Ifo	44841	1990	178384	40.18	0.268
Africa	Kenya	Nairobi	Dagahaley	39441	1990	157784	40.11	0.91

comments	Population	Continent 2	Country 2	administrative level1	name1	administrative unit2
Area	26022188	Africa	Algeria	Department	5	no information
Refugee camp	348311	Africa	Djibouti	District	2	Subdivision
Refugee camp	348311	Africa	Djibouti	District	2	Subdivision
Refugee camp	348311	Africa	Djibouti	District	2	Subdivision
Refugee camp	348311	Africa	Djibouti	District	3	Subdivision
Refugee camp	53191127	Africa	Ethiopia	Province	7	no information
Refugee camp	53191127	Africa	Ethiopia	Province	7	no information
Refugee camp	53191127	Africa	Ethiopia	Province	7	no information
Refugee camp	53191127	Africa	Ethiopia	Province	7	no information
Refugee camp	25241987	Africa	Kenya	Province	4	District
Refugee camp	25241987	Africa	Kenya	Province	4	District
Refugee camp	25241987	Africa	Kenya	Province	4	District
Refugee camp	25241987	Africa	Kenya	Province	4	District

name2	aire unit sqkm	perimeter unit km	Country 2 2	type of climate
0	384780.75	3383.81	Algeria	warm subtropics (winter rain)
1	2317.15	188.419	Djibouti	warm
1	2317.15	188.419	Djibouti	warm
1	2317.15	188.419	Djibouti	warm
5	1911.73	204.362	Djibouti	warm
0	284228.27	2722.31	Somalia	warm
0	284228.27	2722.31	Somalia	warm
0	284228.27	2722.31	Somalia	warm
0	284228.27	2722.31	Somalia	warm
22	27884.43	787.189	Kenya	warm
6	45589.18	1134.78	Kenya	warm
6	45589.18	1134.78	Kenya	warm
39	57727.33	1082.88	Kenya	warm

Fig 6

Length of Growing Period	AREA km2	PERIMETER k	AGECID	IND1	POP	POP L	POP I	AREA HA2	PRES DENS	POP DEN L
0 days (warm)	1430856.19	5948.49	122	00	518220	868438	868438	138180000	0	0
0 days (warm)	22571.78	835.188	1305	10	0	0	0	0	0.03	0.01
0 days (warm)	22571.78	835.188	1305	10	0	0	0	0	0.03	0.01
0 days (warm)	22571.78	835.188	1305	10	0	0	0	0	0.03	0.01
0 days (warm)	22571.78	835.188	1305	10	0	0	0	0	0.03	0.01
0 days (warm)	108958.78	1451.02	1308	21	2921900	0	0	17589000	0.05	0.01
0 days (warm)	108958.78	1451.02	1308	21	2921900	0	0	17589000	0.05	0.01
0 days (warm)	108958.78	1451.02	1308	21	2921900	0	0	17589000	0.05	0.01
0 days (warm)	87235.41	1848.55	1480	21	2921900	0	0	17589000	0.05	0.01
0 days (warm)	13085.44	540.878	1052	00	194240	0	0	8781000	0.03	0.01
0 days (warm)	50277.43	858.934	1052	00	194240	0	0	8781000	0.03	0.01
0 days (warm)	50277.43	858.934	1052	00	194240	0	0	8781000	0.03	0.01
0 days (warm)	50277.43	858.934	1052	00	194240	0	0	8781000	0.03	0.01

Continent	country	Capital	camps	refugee	year	Poptimesyear	longitude	latitude
Africa	Ethiopia	Addis Ababa	Kelalo	0:	0:	0:	44.3:	5.7:
Africa	Ethiopia	Addis Ababa	Mustahil	0:	0:	0:	44.8:	5.2:
Africa	Ethiopia	Addis Ababa	Dolo	0:	0:	0:	42:	4.2:
Africa	Ethiopia	Addis Ababa	Sufte	0:	0:	0:	41.2:	4.3:
Africa	Kenya	Nairobi	Liboi	48478:	1990:	193904:	41:	0.368:
Africa	Kenya	Nairobi	Kakuma	29397:	1990:	117588:	35.24:	3.49:
Africa	Kenya	Nairobi	Mandera	38900:	1990:	155800:	41.3:	3.62:
Africa	Kenya	Nairobi	Thika	4700:	1990:	18800:	36.32:	-0.99:
Africa	Kenya	Nairobi	Banissa	18000:	1990:	72000:	40.67:	3.92:
Africa	Mauritania	Nouakchott	Fassala	15000:	1991:	45000:	-5.68:	15.71:
Africa	Mauritania	Nouakchott	Aghor	15000:	1991:	45000:	-5.99:	18.4:
Africa	Mauritania	Nouakchott	Bassakounu	15000:	1991:	45000:	-6.35:	16.01:
Africa	Niger	Niamey	Nguigmi	2500:	0:	0:	13.07:	14.15:

comments	Population	Continent 2	Country 2	administrative level1	name1	administrative unit2
Reception center	53191127:	Africa	Ethiopia	Province	7	no information
Reception center	53191127:	Africa	Ethiopia	Province	7	no information
Reception center	53191127:	Africa	Ethiopia	Province	11	no information
Reception center	53191127:	Africa	Ethiopia	Province	11	no information
Refugee camp	25241987:	Africa	Kenya	Province	4	District
Refugee camp	25241987:	Africa	Kenya	Province	8	District
Refugee camp	25241987:	Africa	Kenya	Province	4	District
Refugee camp	25241987:	Africa	Kenya	Province	8	District
Refugee camp	25241987:	Africa	Kenya	Province	4	District
Refugee settlement	1995755:	Africa	Mauritania	Region	7	no information
Refugee settlement	1995755:	Africa	Mauritania	Region	7	no information
Refugee settlement	1995755:	Africa	Mauritania	Region	7	no information
	8154145:	Africa	Niger	Department	2	Arrondissement

name2	aire unit sqkm	perimeter unit km	Country 2 2	type of climate
0	284228.27	2722.31	Ethiopia	warm
0	284228.27	2722.31	Ethiopia	warm
0	118103.07	1986.88	Ethiopia	warm
0	118103.07	1986.88	Ethiopia	warm
8	45589.18	1134.78	Kenya	warm
37	82854.92	1403.87	Kenya	warm
22	27884.43	787.189	Kenya	warm
28	7497.75	577.302	Kenya	warm/moderately cool tropics
22	27884.43	787.189	Kenya	warm
0	188982.78	2078.85	Mauritania	warm
0	188982.78	2078.85	Mauritania	warm
0	188982.78	2078.85	Mauritania	warm
28	124832.52	1515.04	Niger	warm

Length of Growing Period	AREA km2	PERIMETER k	AGECID	IND1	POP	POP L	POP I	AREA HAZ	PRES DENS	POP DEN L
1-74 days	178413.81:	3403.4:	1309:01		4803200:	437990:	850852:	28329000:	0:	0:
1-74 days	178413.81:	3403.4:	1309:01		4803200:	437990:	850852:	28329000:	0:	0:
1-74 days	178413.81:	3403.4:	1309:01		4803200:	437990:	850852:	28329000:	0:	0:
1-74 days	178413.81:	3403.4:	1309:01		4803200:	437990:	850852:	28329000:	0:	0:
1-74 days	191823.88:	3991.87:	1057:01		4112300:	403982:	792562:	32342000:	0:	0:
1-74 days	191823.88:	3991.87:	1057:01		4112300:	403982:	792562:	32342000:	0:	0:
1-74 days	125214.23:	2089.07:	1057:01		4112300:	403982:	792562:	32342000:	0:	0:
1-74 days	1070.88:	220.899:	1035:01		145490:	83890:	77318:	912000:	0:	0:
1-74 days	125214.23:	2089.07:	1057:01		4112300:	403982:	792562:	32342000:	0:	0:
1-74 days	489683.71:	3585.7:	129:01		1014900:	217793:	443081:	48844000:	0:	0:
1-74 days	489683.71:	3585.7:	129:01		1014900:	217793:	443081:	48844000:	0:	0:
1-74 days	489683.71:	3585.7:	129:01		1014900:	217793:	443081:	48844000:	0:	0:
1-74 days	504882.79	3987.44	151:00		1002500	409987	798371	52554000	0	0:

Among these problems, the ones that show up repeatedly in the UNHCR bibliography and which seem to be the most crucial are those linked to deforestation.

Malawi, for example, having admitted almost 1 million Mozambicans into their national population of 10 million, suffered from grave environmental problems, especially those resulting from deforestation around refugee settlements (CURE 1993).

Figure: 7 shows the location of all protected areas in Malawi according to their degree of protection.

This large influx of refugees causes other environmental and socio-economic problems as well such as: soil erosion, desertification, and/or problems concerning relations between local and refugee populations.

As underlined by R. Blake and J. Kamangira, it is important not to systematically generalize that refugees always have a negative effect on the environment. Actually, in Malawi, "the large influx of refugees served only to highlight, and then to exacerbate, the already existing problems" (J. Kamangira 1993).

The notion of early warning (T. Dimitrichev, 1991), in environmental terms, passes through the integration of demographic information linked to a country's own administrative sub-units as mentioned by U. Deichmann in: "Medium resolution population database for Africa: Deichmann U., University of California, Santa Barbara, 1994.

In this regard, the World Bank Guidelines drawn up by Cernea (1988) suggest that particular care should be taken concerning environmental matters where displaced people make up more than 10% of the population in the host area.

On the other hand, knowing the population density for any administrative sub-unit, is an essential parameter for the planning phases, notably concerning the availability of land (personal communication Mr Thiadens).

In order to illustrate this problem, we have used the files relative to the protected zones (IUCN, WCMC) and to "wilderness" zones (Sierra Club, World Bank).

The same step will be applied to tropical forests as soon as we have included this "layer" coming from the WCMC into the database.

We have also conceived the database to find out which sites are located inside or on the borders of protected areas and wilderness. This example highlights in particular **the importance of reliable geographic information** concerning the position of refugee sites. A poor location will surely involve erroneous information. We are also waiting for feedback from the Field Officers in charge of the selected camps (Uganda fax of 7 June ref: 590.GEN).

PROTECTED AND WILDERNESS AREAS IN MALAWI

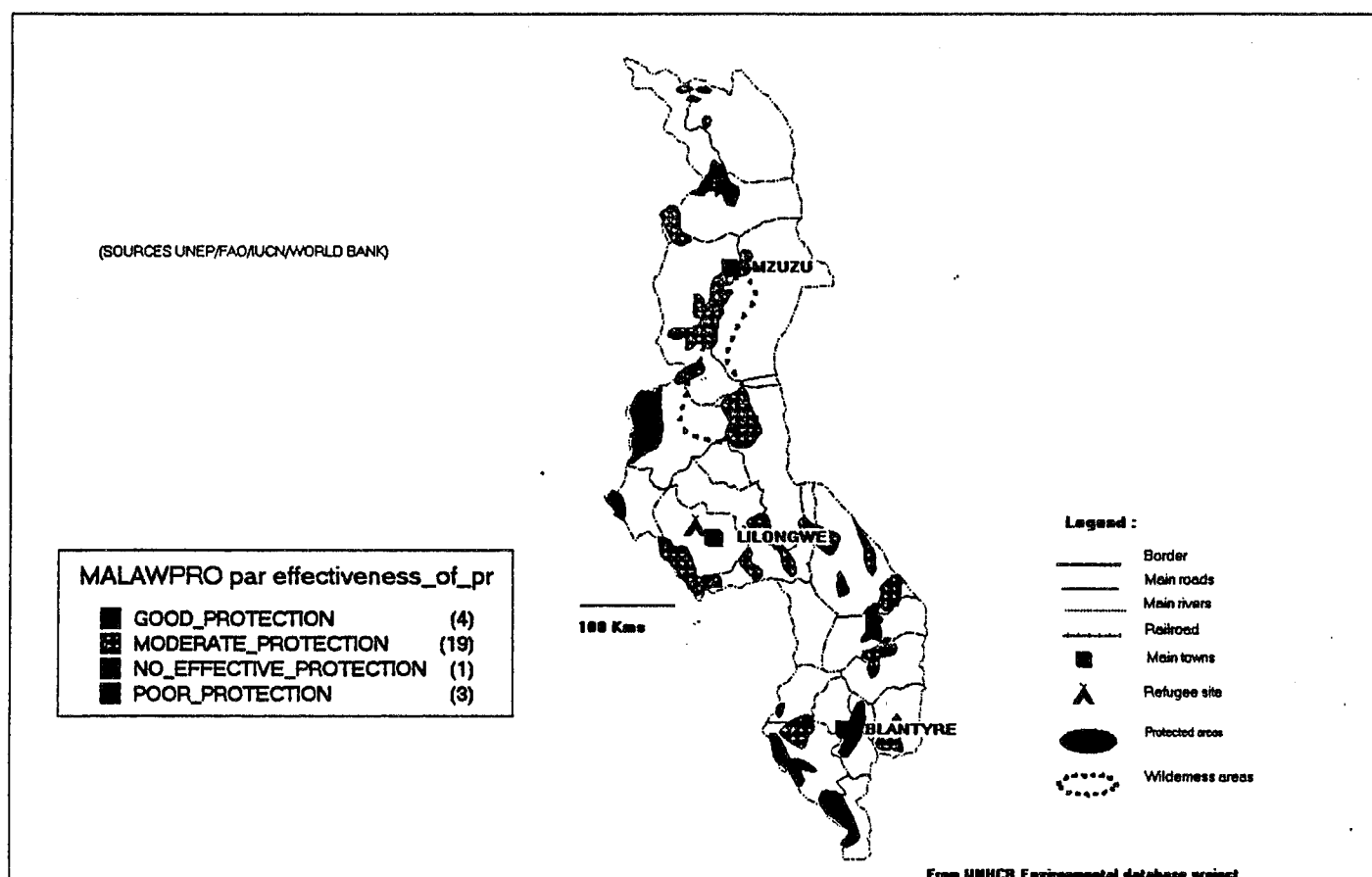


Fig 7

Type of Feature	Number of sites selected
Wilderness areas	7 (listing fig n°8)
Protected areas	12

Table 3: list of selected sites located within or near protected or wilderness areas

Figures 9 and 9 bis present a regional cartography of refugee sites located within protected zones as well as the list of all sites concerned in Africa.

A particular example was created for Uganda in order to show how the GIS can concretely intervene within a planning and prevention policy.

Figure 10 indicates the number of sites located in newly established sensitive zones which are situated 10 to 20 km from a protected area.

This technique, called "Buffer", can be applied in many planning and protection operations (creation of a buffer zone in order to define land-mined areas, i.e. 10 km radius from airport "X" etc.). This is essential when one wishes to carry out protection operations or even logistical operations (distance of the site to a supply source).

In table 4, we have the following results for Uganda:

Number of site selected	type of Buffer zone
1 site	no buffer (protected area)
2 sites	10 km buffer
4 sites	20 km buffer

Table 4: Number of selected refugee sites, according to given buffer areas of 10 and 20km

Within GIS, we can also use the already existing indexes to measure the impact on a country's natural resources by a population of refugees.

Among these, the World Resources Institute (WRI) provides diskettes that show people how to calculate "relative refugee absorption capacity" by country and based on natural resources variables. The formula takes into account the following parameters: Technology, Population Stability, Land Resources, Water Availability, Food Self-Sufficiency, and Forest Conservation.

According to these relative indexes, this will be very useful, for example, to estimate at what rate of deforestation UNHCR must consider in the importation of energy supplies and what kind of strategy it must adopt (J. Dennis, 1993). Obviously, all this information must be validated in the field.

Continent	Country	Capital	camps	number of refugee	year of creation	date of information	longitude
Africa	Mauritania	Nouakchott	Aghor	15000	1991	01/12/1993	-5.99
Africa	Mauritania	Nouakchott	Fassala	15000	1991	01/12/1993	-5.68
Africa	Kenya	Nairobi	Kakuma	29397	1990	01/12/1993	35.24
Africa	Ethiopia	Addis Ababa	Dharwanaji	30000	1990	01/12/1993	42.94
Africa	Ethiopia	Addis Ababa	Fugnido	19587	1990	01/12/1993	34.58
Africa	Uganda	Kampala	Ogubebe	0	0	01/12/1993	31.8
Africa	Lesotho	Maseru		0	0	01/12/1993	28.2

latitude	type of camps	administrative level1	name1	administrative unit2	name2	aire unit sqkm
16.4	Refugee settlement	Region	7	no information	0	168982.78
15.71	Refugee settlement	Region	7	no information	0	168982.78
3.48	Refugee camp	Province	8	District	37	62854.92
8.97	Refugee camp	Province	7	no information	0	284228.27
7.78	Refugee camp	Province	8	no information	0	41843.84
3	Reception center	Region	3	District	11	5243.77
-29.4	Urban refugees	District	5	no information	0	8401.77

Fig 8

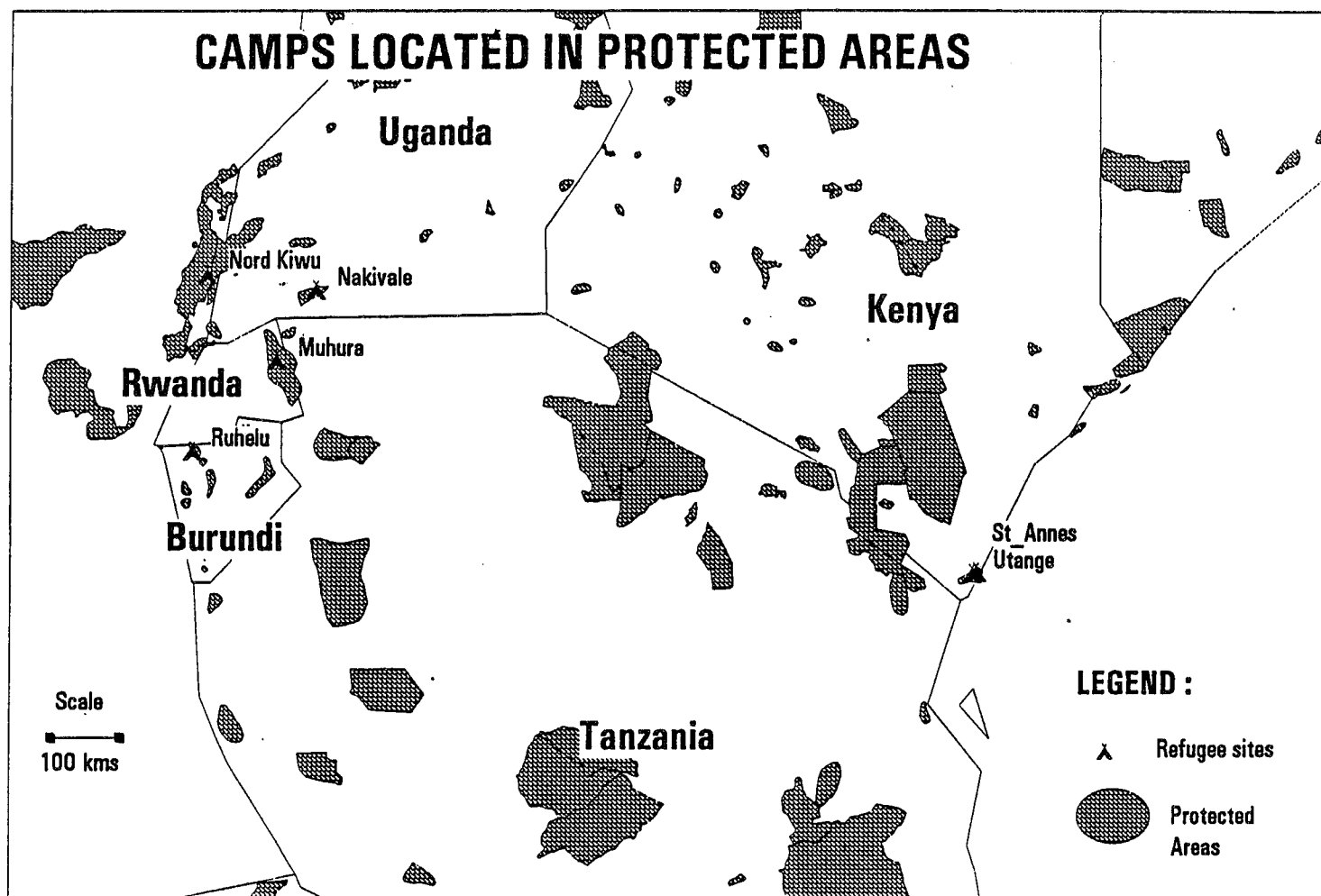


Fig 9

Continent	Country	Capital	camps	number of refugee	year of creation	date of information	longitude
Africa	Ethiopia	Addis Ababa	Bonga	13757	1990	01/12/1993	34.37
Africa	Central_Africa	Bangui	Mboki	30000	1990	01/12/1993	28.1
Africa	Uganda	Kampala	Nakivale	0	0	01/12/1993	31
Africa	Zaire	Kinshasa	Nord Kiwu	1500	1990	01/12/1993	29.81
Africa	Rwanda	Kigali	Ruhoku	10838	0	01/12/1993	29.41
Africa	Rwanda	Kigali	Muhura	0	0	01/12/1993	30.5
Africa	Kenya	Nairobi	Utange	42381	1990	01/12/1993	39.35
Africa	Kenya	Nairobi	St Annes	0	0	01/12/1993	39.4
Africa	Zambia	Lusaka	Ukwim	27000	1990	01/12/1993	30.54
Africa	Namibia	Windhoek	Osiri	1000	1992	01/12/1993	17.19
Africa	South_Africa	Pretoria	Pafuri	50000	0	01/12/1993	31.03
Africa	South_Africa	Pretoria	Olifants	60000	0	01/12/1993	31.31

latitude	type of camps	administrative level1	name1	administrative unit2	name2	aire unit sqkm
8.18	Refugee camp	Province	8	no information	0	41843.84
5.2	Refugee camp	Prefecture	3	no information	0	58035.98
-0.7	Refugee settlement	Region	4	District	2	18819.94
-0.5	Area	Region	8	no information	0	250445.31
-2.78	Refugee camps	Prefecture	4	no information	0	2615.48
-1.8	Refugee camp	Prefecture	7	no information	0	5836.1
-4.32	Refugee camp	Province	2	District	18	8884.28
-4.3	Reception center	Province	2	District	18	8884.28
-14.18	Refugee settlement	Province	3	District	48	18724.77
-20.59	Refugee camp	no information	0	no information	0	815712.7
-22.54		Bantustan no-Indept	13	Statistical region	58	87864.68
-23.7		Bantustan no-Indept	13	Statistical region	58	87864.68

AU HA	FIN AU HA	IND2	POP DEN	CLASS	AREA	PERIMETER	D ITEM	Country 2	NAME
0.37	0.37	0	2	1	0.858	4.788	634	Ethiopia	GAMBELLA
0.01	0.01	0	2	1	9.355	27.228	727	Zaire	GWANE
0.247	0.247	0	8	1	0.901	4.881	1	Uganda	LAKE MBURO
0	0	0	0	0	0.221	2.745	0	Zaire	VIRUNGA
0.318	0.357	0	8	1	2.118	12.474	1036	Rwanda	KIBIRA I
0.084	0.084	0	3	1	0.498	3.043	1042	Rwanda	AKAGERA
0.252	0.252	0	4	1	0.423	3.894	1129	Kenya	SHIMBA HILLS
0.252	0.252	0	4	1	0.423	3.894	1129	Kenya	SHIMBA HILLS
0.048	0.048	0	2	1	7.992	24.055	1370	Zambia	WEST PETAUKE GMA
0.087	0.087	0	2	1	18.704	37.107	1482	Namibia	WATERBERG PLATEAU
0	1.174	1	1	1	1.829	7.372	1538	South_Africa	KRUGER NP
0	1.174	1	1	1	1.829	7.372	1538	South_Africa	KRUGER NP

effectiveness of protection	AREA 3	PERIMETER 3 SIZE	REAL AREA	OLDID	CODE
no effective protection	8970.88	336.454	2000	6308.79	160 99
no effective protection	4130.08	354.877	0	3659.85	235 482
moderate protection	588.824	111.339	538	523.14	342 199
no effective protection	7545.34	948.55	7800	8908.08	314 511
poor protection	233.997	60.5988	379	214.98	378 228
poor protection	2852.53	261.575	2500	2821.11	352 208
moderate protection	432.032	89.2347	192	429.17	398 242
moderate protection	432.032	89.2347	192	429.17	398 242
poor protection	3888.38	348.57	4140	3947.51	473 288
moderate protection	888.428	145.528	405	901.07	581 588
good protection	19188.74	1015.3	19485	19532.58	578 575
good protection	19188.74	1015.3	19485	19532.58	578 575

Fig 9 bis

The analyses of requests linked to environmental questions (and possibly to other questions) can be multiplied and begun rather quickly, considering the quantity of information available within the database. In this sense, the GIS and the database show, if we rely on the preceding examples, the operational character of the tool during the formulation of precise requests.

It is obvious that complex requests must all rely on qualitative models or quantitative recognized models.

So if one wants to know the risks of soil erosion, even in an empirical and qualitative manner, it is advisable to use USLE (Universal Soil Loss Erosion, Whismeyer and Smith 1965). To elaborate this model we must have precise information, for example, on soils, landuse and slopes.

For Africa, UNEP has developed global vulnerability maps in different areas such as: soil degradation, typology of drylands, water erosion, wind erosion and desertification (UNEP/GRID sources). They will be directly integrated into the database.

If more precise studies must be undertaken (on a larger scale), a systematic research of pertinent information must be considered, associated to the selected model, by using the capacities of the GIS.

At this level, the building-up of a database used for this kind of research, calls for a long and important amount of work (collecting information, digitalization of information, processing).

5.4.2 Evaluation

At this stage of the study, the database does not allow us to carry out the evaluation of environmental degradation caused by the establishment of refugee camps.

Information relative to the evaluation of "degradation" is only accessible by the reports written during consultations or by joint projects between UNHCR and the organizations involved (UN organizations, NGO's, bilateral agencies, etc.).

As we have already discussed, deforestation is a crucial problem (S. Gurman, Y. Stevens, CURE, K. Watanabe, etc.). In many countries, UNHCR is involved in projects linked to this phenomenon (Pakistan, Iran, Kenya, Malawi, etc.).

In this field, the GIS can help us at two levels:

1. The first concerns identification (cf preceding paragraph).

In addition to the remarks made in the preceding paragraph, the GIS also allows us to create "scenarios". We can, as shown in figure 10, actually establish a typology of potential vulnerability according to buffer zones. We can assume a

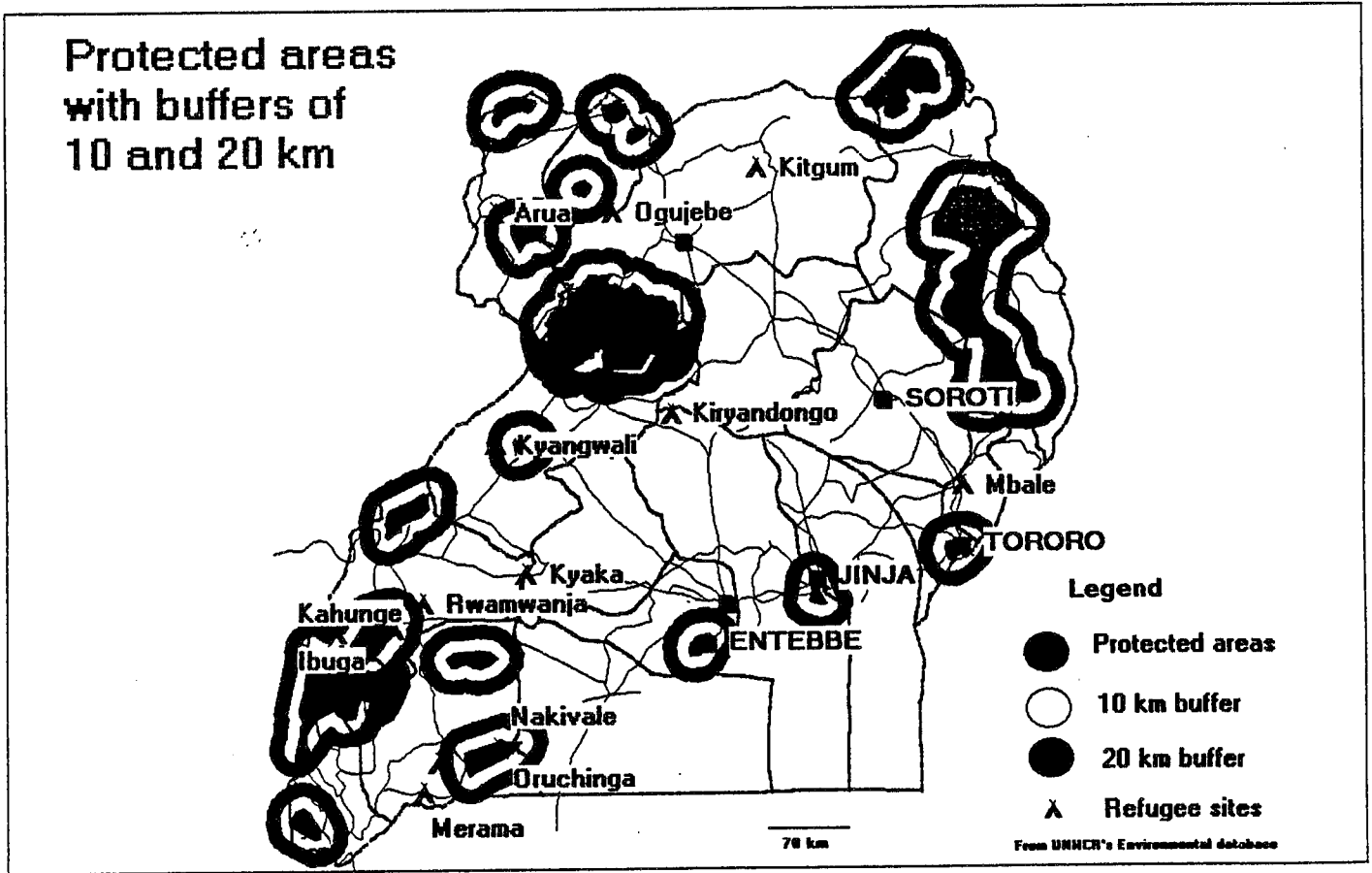


Fig 10

high sensitivity index for sites located at a distance of less than 10 km from a forest or a protected zone and a weaker sensitivity index for sites located at a distance greater than 20 km. The requests may become more and more complex.

They may call for several criteria of the same parameter, or call for several different parameters, i.e. selection of sites located within a protected zone (option: no protection), up to a distance of 10 km from the zone.

Figure 11 shows all of the refugee sites and the principal localities situated at a distance of 50 km from the Rwandan border.

2. The second concerns estimation.

In collaboration with competent operators, UNHCR may then decide to set up projects aimed at the protection, management and rehabilitation of affected zones (Klingshirn et al 1992).

Within this specific domain (deforestation), the GIS proves to be an essential tool. Linked to the GPS (Global Positioning System cf appendix 2) and to remote sensing, it enables a certain number of analyses to be done.

Leading studies over the past several years have shown that satellite imagery was (all conditions being otherwise equal) a tool which performed equally well, in terms of productivity and forecasting, in the evaluation of cleared land surfaces (due to deforestation) as in the management of forests.

Remote sensing, linked to GIS technology, has contributed to much progress in various fields of research in the domain of deforestation. Many studies give evidence of this progress: Deforestation: Sader 1990, Shreier 1989, Stone 1991, Allan UNEP/GRID 1990, Resource Management: Olsson 1986.

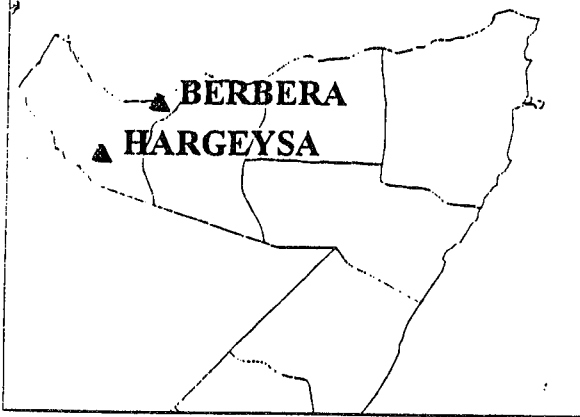
The GPS allows, notably :

- to specify the geographic location of all points on the surface of the globe (cf appendix 2 and fig. 12);
- to adjust the satellite images in order to geometrically correct them (location of landmark points); and
- to measure (perimeters) objects situated on the earth's surface.

One can also use remote sensing to analyze the evolution of land-use, especially the destruction of certain vegetation (grasslands). However, obtaining information on the evolution of the landscape from satellite imagery remains difficult without a serious ground "truthing". Actually, landscapes of third world countries have transition limits which are relatively blurry (as opposed to forests).

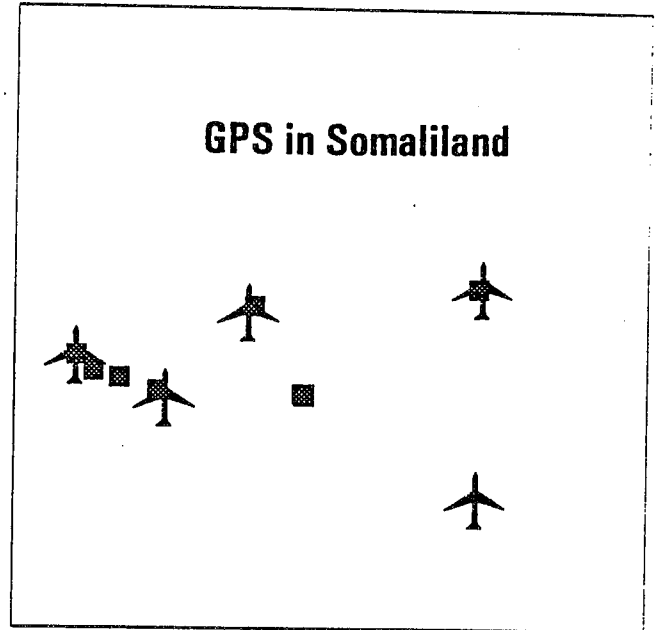
Initial Map Somaliland (Somalia)

1



GPS in Somaliland

2



GPS in Somaliland Towns and Airports Locations with GPS technology

3

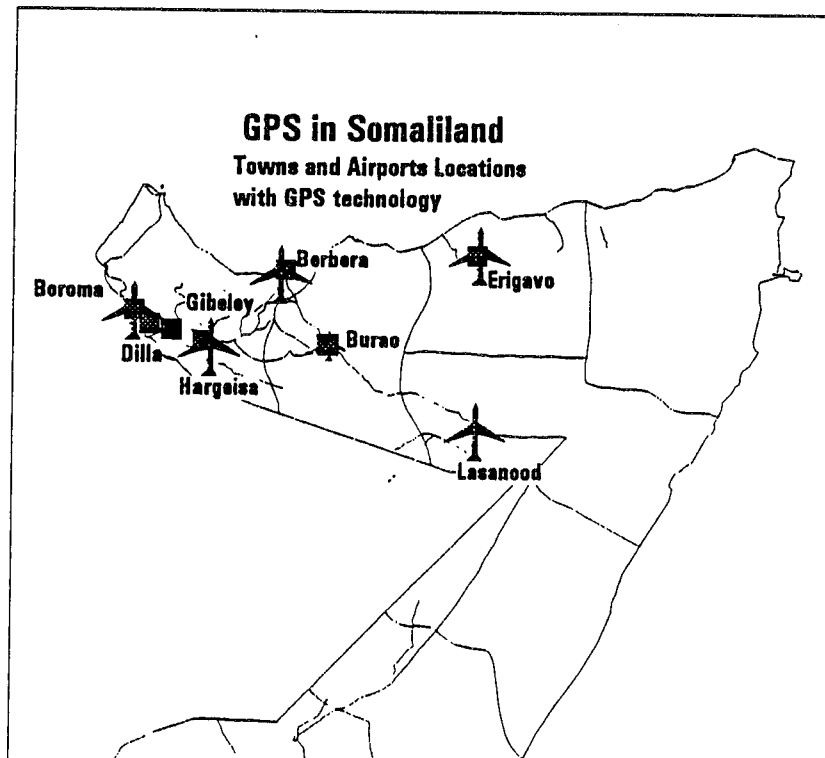


Fig 12

<u>BOROMA</u>		
Town	N 09° 56'.000 E 43° 10'.750	-9.93 / 43, ¹²⁶ 1875
Airport	N 09° 56.600 E 43° 09'.300	-9.95 / 43.233
<u>HARGEISA</u>		
Town	N 09° 34'.500 E 44° 03'.500	-9.705 / 44, 188.
Airport	N 09° 30'.877 E 44° 05'.085 (par GPS)	-9.7436 / 44.1069
<u>BURAO</u>		
Town	N 09° 31'.750 E 45° 32'.000	-9.725 / 45.533
Airport	N 09° 31'.087 E 45° 33'.870 (par GPS)	-9.5408 / 45.7916
School	N 09° 32'.115 E 45° 32'.425 (par GPS)	-9.5652 / 45.6518
Pump	N 09° 32'.329 E 45° 31'.820 (par GPS)	-9.66247 / 45.744
<u>BERBERA</u>		
Town	N 10° 26'.500 E 45° 00'.000 10.43/	-10.5722 / 45.0166
Airport	N 10° 23'.767 E 44° 57'.162 (par GPS) 10.38.	-10.5963 / 44.995x
<u>LAS ANOOD</u>		
Airport	N 08° 29' 10 E 47° 20'.000	-8.4833 / 47.333
<u>ERIGAVO</u>		
Town	N 10° 37'.000 E 47° 21'.750	-10.6166 / 47.5583
Airport	N 10° 38' 586 E 47° 23'.154 (par GPS)	-10.7961 / 47.4261
<u>DILLA</u>		
Town	N 09° 46'.500 E 43° 21'.000	-9.9055 / 43.35
<u>SIBELEY</u>		
Town	N 09° 42'.000 E 43° 37'.250	-9.7 / 43.6861

Because of this, it is difficult to define precisely (the period of evolution of the landscape is too long compared to the rather recent development of satellite technology/multi-date analysis), within a reasonable margin of error, the development of certain types of land-use (J.Y. Bouchardy, L. Guyot, 1993).

5.4.3 Prevention

Example 3: Elaboration of decision-making maps for planning

As it is better "to prevent than to cure", it is important, within the framework of an environmental policy aimed at providing lasting solutions, to try new methods, methods which can aid in decision-making while at the same time limiting the negative aspects.

Concerning environmental questions linked to refugees, it is obvious that emergency situations very rarely allow a thorough evaluation of whether a potential site presents the best possible conditions for the refugees (Y. Stevens, J. Kamangira). Otherwise, most often the host country decides on a spot where they would like to see these populations go (personal communication: R. Thiadens).

It is important, however -- even if the opportunities are few --, to have documents in hand which may, especially when discussing the choice of a site with local authorities, foster public awareness in this domain. These cartographic documents must support the discussions and place those accountable for the UNHCR in a position where they can use these documents to steer decision-making.

It is essential to try to make the local authorities understand that wise planning -- by limiting certain ill-fated components of the following parameters: ethnic, economic and environmental -- is an essential endeavour for all parties involved: refugee population, local population, host country and UNHCR.

To use just one example concerning the environmental questions linked to refugees, the choice of a "suitable" site should theoretically allow :

- | | |
|-------------------------------|--|
| <u>for UNHCR:</u> | a reduction in costs concerning the rehabilitation of sites affected by a refugee population; |
| <u>for the host country :</u> | a better management of natural resources by preserving its environment in the form of lasting development. |

Coming back to the idea developed by J. Dennis on the zones more or less favorable to the establishment of new sites (EESARHA model, cf appendix 5), the GIS and the related cartography which follows from it can facilitate this approach.

In this sense we can propose for a given country a synthesized cartography showing a certain number of regions where implantation of a site would be :

- either harmful for the lives of the refugees (i.e. type of unfavorable climate, etc.);
or
- harmful to the environment (i.e. proximity to a protected zone, etc.)

The searches and the complexity of the maps can, of course, be increased, notably by selecting zones close to airports, located at a distance "x" from an access runway or at a distance "y" from a reservoir, buffer zone, etc.

This cartographic approach must underline the zones which, according to certain parameters, present restrictions concerning environmental questions.

For the time being not working on the same scale as the physical planners (who most often work at 250000°, a cartographic scale common in Africa), certain essential parameters such as the relative flat area of terrains (UNHCR Emergency Handbook) are difficult to provide.

Considering our scale of work and the parameters that we have, we can help in the planning of sites by providing situation-specific maps based on the zone (climate, forests, parks, etc.) which can highlight risks concerning the environment.

When referring to the Manual of Emergency Situations (UNHCR 1982), the geographic criteria to apply when choosing a site are the following:

- 1 a sufficient supply of water throughout the length of the year (this is the base criterion);
- 2 topography and drainage (a flat enough and well-drained region);
- 3 the surface (WHO advocates 30 square meters per person);
- 4 the socio-cultural background of the refugees;
- 5 ease of access (proximity to means of communications);
- 6 the environment is considered in the work in relation to endemic diseases;

- 7 the nature of the soil (conditions favorable to agriculture);
- 8 land-use -- the terrain must possess a good vegetation cover --, which provides shade and limits soil erosion; and
- 9 the basic rights (vacant terrain).

Of course, we strive to optimize a site according to these criteria but in reality this remains very difficult (such as the urgency of the situation, the political will of the host country, etc.).

Regarding the recommendations made by the manual, the proposed database can already quickly provide general information concerning all the countries of Africa.

These parameters are (for the moment):

- the typology of surface communication means;
- the typology of rivers;
- the main cities;
- the main airports and aerodromes;
- regions which are very uneven from a topographical point of view;
- the population density of the countries;
- soil map (we are awaiting its arrival);
- the numerous types of land-use (we are awaiting its arrival);
- the map of reservoirs and of small lakes.

Added to this base information, environmental information can of course be included. If the engineers in charge of establishing sites generally work at 250000°, in certain cases up to 50,000°, the first work to be done is to evaluate the "target" regions that are potentially to be found in a general manner.

In this context, the elaboration of general information maps presented above constitute an essential and irreplaceable source of information in the first phases of analysis.

Concerning environmental questions and UNHCR policy, the establishment of decision-aiding maps presented earlier can obviously be one of the essential components of this policy.

In the following example (fig. 13), we demonstrate an application of this for Kenya. In this map, besides the cited base information, we have underlined the regions which from a climatic point of view expose critical thresholds for refugees and elsewhere and the environmental zones where it is advisable to avoid when establishing a site.

VI. CONCLUSIONS AND RECOMMENDATIONS

This first phase of the project has been completed.

A certain number of information has already been integrated into the database.

The database contains 239 refugee sites found throughout the world, 169 of which are in the African continent alone.

The first results concerning the identification of refugee sites, considering the integrated environmental parameters, already form an innovative and non-negligible basis of thought for an operational environmental policy within the UNHCR.

In fact, the establishment of "environmental" programs in collaboration with other partners, happens first of all through a good knowledge of the ecological context of the refugee sites.

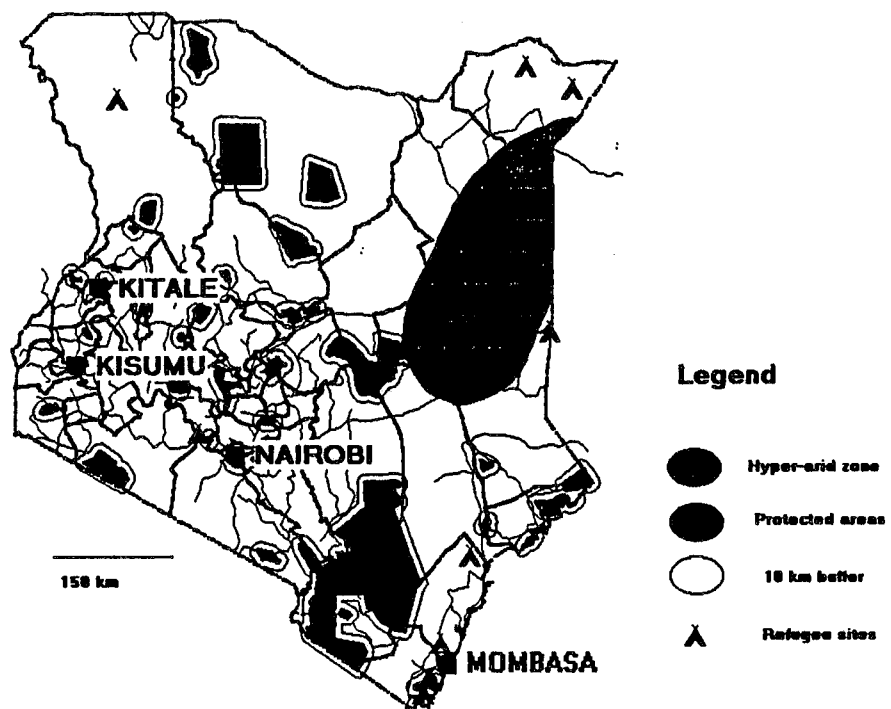
The database provides an "**environmental identity card**" of sites across the world, in a systematic and renewable manner. Then from this "identity card" a flexible and adaptable methodology will follow (calling for information verification, notably on the ground with the Field Officers) which should engage plans of action in a more operational and systematic manner, according to the degree of "gravity" measured.

Obviously, this estimation must be increased and refined according to new environmental parameters which must increase the GIS database.

The increase and the systematic collection of environmental information within the database should allow an approach to the recommendations formulated for the establishment of the E.A. (Environmental Assessment) (Dennis J., Gurman S.) (cf Appendix 5).

This tool is open to all the services of UNHCR, it allows for continual updating and affords those at UNHCR in charge of the management and the planning of refugee sites a constantly renewable source of information which should quickly become indispensable.

Decision Map for Kenya taking in account Protected areas and Hyper-arid zone



From UNHCR's Environmental database

Fig 13

Of course, this tool and the associated database are open to all services of UNHCR. They notably allow for the monitoring in real time of emergency situations (i.e. Rwanda) by offering at any moment:

- an instantaneous localization map (the state of sites in real time), which can be quickly updated by new geographical information coming from the field (cf Appendix 7 Emergency Cartography);
- a regional vision of refugee sites (cf fig 14);
- a logistical analysis of the region, according to parameters such as: types of roads, cities, airports, distance from the site to closest city and airport, etc.

The advantage of a GIS-type tool within UNHCR becomes more and more clear, notably for the cartographic plan. Several sections have already called upon this tool in different fields: PTSS: Environment and Cartography; Program Policy Unit, Desk Officers (Rwanda emergency unit), P.I., Food and Statistical Unit: Cartography and Logistics.

At this stage of the project, the recommendations that we can give UNHCR are the following:

A/ At the environmental database level

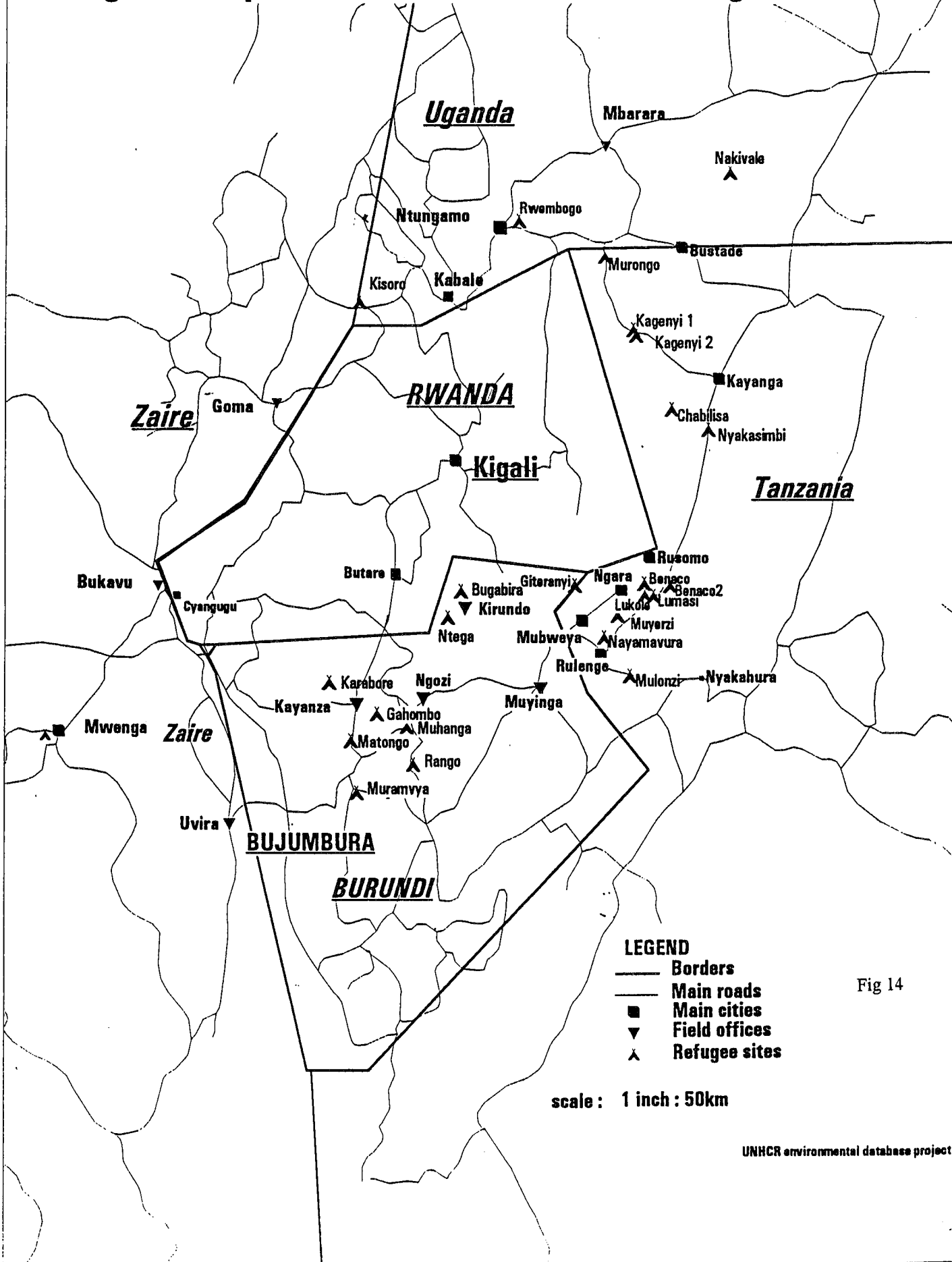
GIS, in its theoretical conception and in its operation, requires precise information regarding the geographic location of objects (refugee sites). One of the first "deficiencies" that we encountered concerns the systematic absence of the exact geographic location of refugee sites (knowledge of the latitude/longitude).

- 1 Considering that the base information of our data bank is the refugee sites, it is thus **urgent to systematically increase the precision linked to the geographic location of all the refugee camps**. This can be easily done notably by integrating into the mission reports the precise location of the sites in latitude/longitude. It is also advisable, as a general rule, **to increase the precision and the reliability of the information concerning refugee sites** (number of refugees, date of the creation of the camps, etc.). The database asserts itself as the tool which can allow this to be fulfilled, in relation with all the other concerned services of the UNHCR.

1.1 For this, it is recommended to systematically increase the collection of cartographic documents (paper and/or numerical) from the different agencies in charge of cartography (cf Appendix IGN). The map remains the essential element in all types of planning.

1.2 From a technical point of view, the geographic co-ordinates can be quickly and easily obtained thanks to the growing use of **GPS**, whose co-ordinates can be directly integrated into GIS (cf Appendix 2, PTSS has already ordered 5 GPS's). These machines (the size of a TV remote

Regional map of main sites for Rwandan refugees



control) allow for the retrieval, in real time, of all geographic points on the surface of the Earth.

Better logistics and better knowledge should follow from this precise localization (notably linked to the sites, i.e. with this perimeter using GIS we can determine the area of the camps and thus know, according to the population: the density, the pressure on the ground, etc.), thus saving time and money.

- 2 In the face of the development of the project and considering the growing interest of the database for various services of the UNHCR, it is recommended to increase as soon as possible certain numerical sources of information to small (1: 1 million) to have a global view for all the sites. Data relative to a medium scale, close to the planning scale (250,000°/100,000°), must only be acquired for case studies. Indeed, a medium-scale or large-scale covering:

- a) exists only very rarely for developing countries;
- b) represents an enormous quantity of information to process;
- c) for the moment, a certain homogeneity must be kept in the scale of analysis and within the database.

At this stage of the study, only certain parameters necessary for planning must be pursued on a greater scale. These essentially are topography (level curves) and the infrastructure (roads, cities, villages, etc.).

- 2.1 A particular effort must be undertaken to obtain socio-economic as well as demographic data (cf fig. 15). This data often comes into play in the elaboration of:

- a) the search for areas in which to establish sites;
- b) an environmental impact model;
- c) in the early-warning notions (WRI, World Bank).

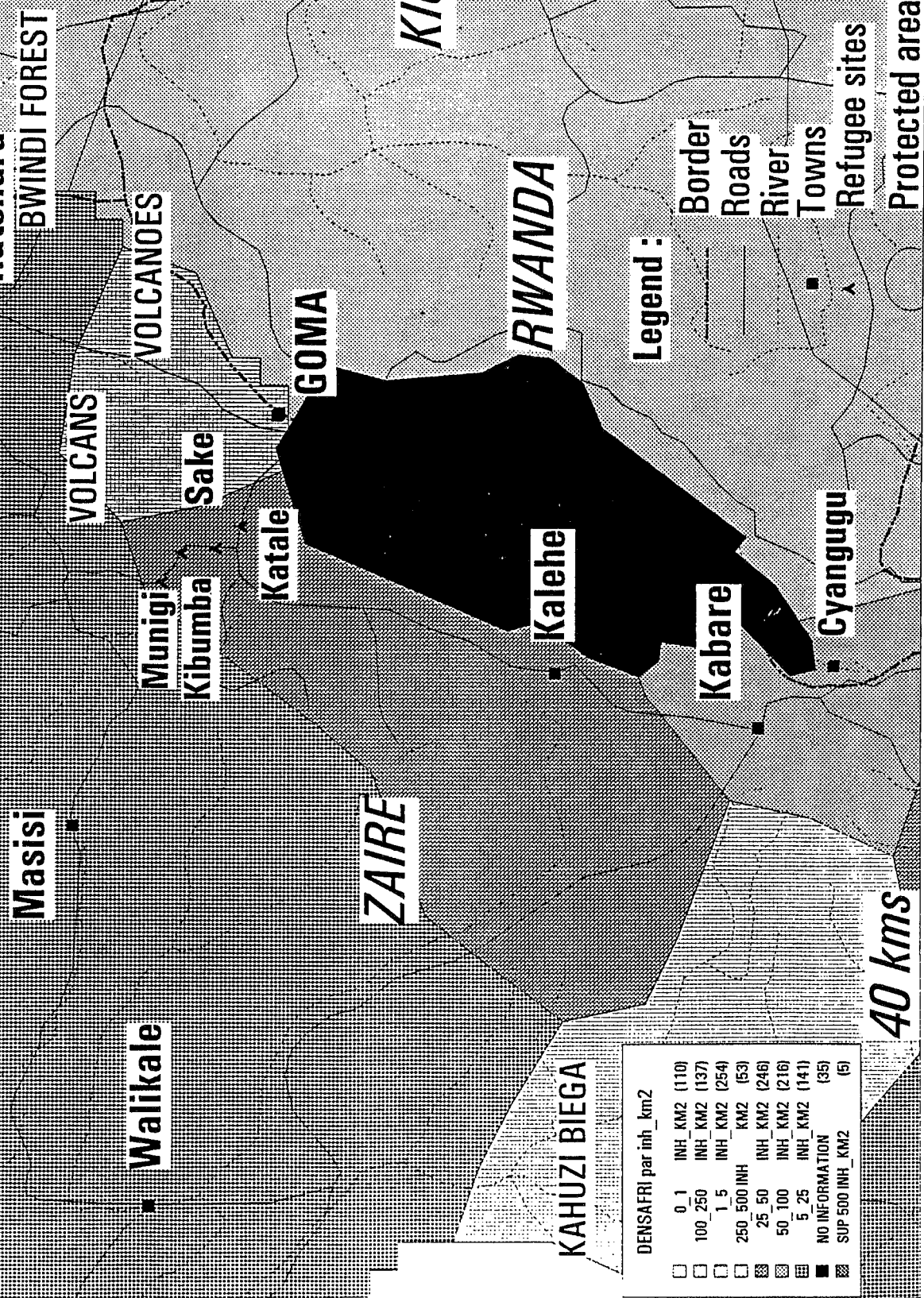
This type of data (tabular) can be easily integrated into the database and then connected to cartographic information (for example, the population per administrative unit).

- 2.2 This collection of information passes notably through the growing use of the **Internet network**.

This telephone network (via the function FTP: File Transfer Protocol, cf Appendix 6) allows quick access to many numerical bases across the world

DENSITIES OF POPULATION ON THE BORDER OF RWANDA AND ZAIRE

Date 20 07 94 (data sources UNEP/FAO)



(Directory of Databases: NASA, UNEP, WRI, Canadian Center for Remote Sensing, USGS, GRASS, NOAA, EROS Data Center, University of California at Santa Barbara, etc.) and to send back in real time, under certain conditions, the selected data (satellite images, digitize maps, etc.). There is no cost to access most FTP sites and no long distance charges either.

2.3 In this context, it is advisable to maintain close cooperation with UNEP/GRID Geneva which provides valuable support during this first phase of the project. (cf M. Stauffacher).

2.4 It is advisable that the UNHCR develop its own database (multi-level and precise) in collaboration with the various agencies and institutions which were contacted or worked with UNHCR (recommendations: J. Dennis, S. Gurman, CURE, R. Blake, PTSS report 91/33 Cambodia).

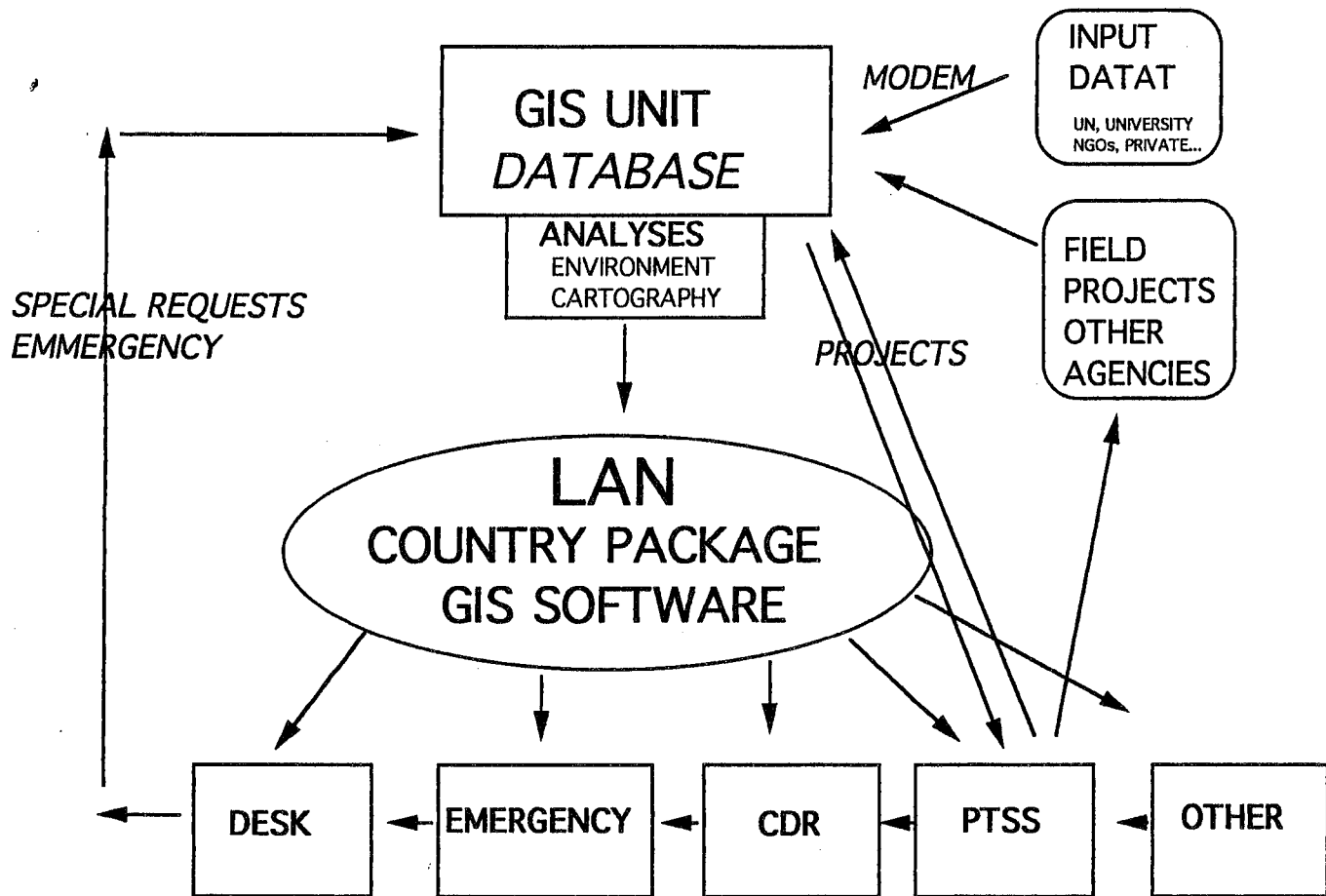
Today, because of important progress in computer science during the past several years, micro-computers now have very rapid processing capacities and increased storage capabilities (optical disk: 1Gbytes). Therefore, these components enable simpler archive storage and a quicker information search.

2.5 Linked to the development of all kinds of information, it is advisable to adopt a software package which is able to recognize raster data (format of satellite image and numerous files, i.e. images from scanners). In June, MapInfo put its latest version on the market which allows integration of raster data.

2.6 This new module (raster) should enable the increased use of satellite images at high resolution (SPOT/TM) in environmental studies, especially those projects linked to deforestation (joint use of software packages IDRISI and MapInfo).

3 It is advisable to systematically develop, when possible, the use of GIS in environmental studies (deforestation, for example, cf recommendations Cambodia, Malawi) lead by UNHCR in collaboration with other agencies or institutions (NGOs, UN organizations etc.). This could also be used for monitoring environmental degradation around refugee sites. Integration and archive storage of projects is then done in UNHCR's general database.

4 In order to use the data from the GIS operationally and systematically, it is advisable, to develop and distribute to all of the Desk Officers and services who wish to have these data: an "environmental and topographic country package" directly accessible via the LAN (cf fig. 16). This package must provide by country: topographical and cartographic data as well as the principal associated environmental components (precipitation, temperatures and type of climate). The basic information remains the refugee site.



PROPOSAL OF A NETWORK FOR UNHCR'S ENVIRONMENTAL DATABASE

Fig 16

- 4.1 In order for this package to be used in an operational manner, it must be very "user friendly". In this regard, it would be desirable to minimize at least the manipulations linked to the use of the software package. Thus, it is advisable to establish an easy-to-use software package on the LAN network.
- 5 UNHCR, in the framework of its environmental policy and via its technical unit (PTSS), must support at a future point the systematic development of a cartography which is helpful in the decision (planning) on where to establish new refugee sites.

This analysis must stress these points:

- for already existing sites: the condition of the location from an environmental standpoint, a hierarchical order of the sensitivity (or vulnerability) of sites;
- for prevention: the establishment of an operational cartography presenting the "sensitive" environmental characteristics (for the environment and for the refugees). This cartography must register as an important element in the development of an environmental policy within the office.

B/ Establishment and Maintenance of the Database

- 1 It is desirable that the database be established in a technical service such as PTSS, this particular service has already been used in numerous environmental projects. The database and the GIS provide a technical activity which is complementary to that of PTSS.
- 2 They must be ensured by a professional who guarantees the research, the management, the integrity and the quality of the information which will in turn be provided to the users (location of refugee sites, map projection use of the GPS, downloading and conversion of data, choice of pertinent data, etc.).

All of these functions, inherent to the smooth operation of the database, call for deeper knowledge in many technical domains, especially cartography, techniques linked to the GIS and GPS, SQL language, use of the electronic network (Internet), knowledge of new products, etc.

- 3 The updating of basic information (the refugee sites) must be done in close collaboration with PCBS (B. Hovi), the Desk Officers and PTSS.

It must be performed (as much as it possibly can) :

- in a "normal" situation: **every year** (January 31 in agreement with the closure of the statistics report). It must be done in phase with the updating done by

the statistics unit in order to avoid multiplying the contradictory sources of information (B. Hovi, PCBS communication).

- in an emergency situation or particular case: **in real time** (via the Desk Officers and PTSS, cf Rwanda).

C/ Links with other sections of the UNHCR and potential uses

There are numerous complementary interests which we were able to see during the development of the project.

They especially concern cartography and, more specifically, integration and linkage of maps into other already existing databases of UNHCR (FOXPRO: PCBS statistics, FOLIOView: especially CDR). In concrete terms, the exchange of information is possible and useful.

The services which are interested are :

1 PCBS

MapInfo enables the integration of many databases (Excel, dBASE etc.). FOXPRO is used by PCBS and allows exportation in dBASE or ASCII format. These two formats are compatible with our database which enables the direct integration of statistics from FOXPRO. MapInfo then allows this statistical information to be linked to cartographic information which is able to underline these parameters.

2 PI/Media/Publications/CDR

Within the activities of these sections, it is possible to integrate data from databases within the systems used by these services.

The exportation of thematic maps from our database can be used (via PCX format/PaintBrush under Windows) as basic information: base map (geographic location of refugee sites and location of principal infrastructures) within the publication work of these units (software package COREL-Draw). The CDR devotes a part of its publications to the geography of countries. In 3 or 4 months, the downloading of maps will be possible with the installation of the new version of FOLIOVIEWS under Windows (personal communication Mr Sabety, Consultant).

3 Emergency Preparedness and Response Section/Supplies and Transport Section

This section by definition must act quickly on the terrain when faced with emergency situations. By taking into consideration the maps from the "UNHCR preparedness profile" and using diverse tools such as: remote sensing and GPS

within the database, it is possible to envision a cartography in "real time" of the intervention sites.

This cartography should facilitate several operations concerning logistics such as: number of airports within a 50 kilometer radius, principal towns, population, communication network, etc. Appendix 7 shows, schematically, the use of these tools within an operational cartography for emergency situations.

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APPENDIX 1**LISTING OF ALL REFUGEE SITES INCLUDED IN THE DATABASE**

Continent	country	Capital	camps	refugee	year	longitude	latitude	comments
Africa	Angola	Luanda	Malanje	11000	1970	16.2	-11.01	Rural sites
Africa	Burkina Faso	Ouagadougou		5000	0	-1.8	12.4	Mixed with local pop
Africa	Burundi	Bujumbura	Muramba	0	0	30	-3	Refugee camp
Africa	Burundi	Bujumbura	Kanyangozi	0	0	30.4	-3.3	Refugee camp
Africa	Burundi	Bujumbura	Kigamba	0	0	30.5	-3.2	Refugee camp
Africa	Burundi	Bujumbura	Mugera Mishiha	0	0	30.6	-3	Refugee camp
Africa	Burundi	Bujumbura	Bukamba	0	0	30	-4.1	Refugee camp
Africa	Burundi	Bujumbura	Kayagoro	0	0	29.8	-4.2	Refugee camp
Africa	Burundi	Bujumbura	Nyaruhengeri	2800	1991	29.95	-2.93	Refugee camp
Africa	Cameroun	Yaounde	Garoua	2000	1980	13.24	9.18	
Africa	Central Africa	Bangui	Boubou	20000	1993	17.47	6.22	Refugee camp
Africa	Central Africa	Bangui	Mboki	30000	1990	26.1	5.2	Refugee camp
Africa	Congo	Brazzaville	Kouilou	1000	1992	12	-4.3	
Africa	Djibouti	Djibouti	Aour Aouassa	7600	0	42.49	11.11	Refugee camp
Africa	Djibouti	Djibouti	Assamo	5300	0	42.82	11.01	Refugee camp
Africa	Djibouti	Djibouti	Holl Holl	7200	0	42.92	11.29	Refugee camp
Africa	Djibouti	Djibouti	Ali Ade	9900	0	42.87	11.14	Refugee camp
Africa	Ethiopia	Addis Ababa	Bonga	13757	1990	34.37	8.18	Refugee camp
Africa	Ethiopia	Addis Ababa	Daror	16000	1990	46.6	7.7	Refugee camp
Africa	Ethiopia	Addis Ababa	Terefi Ber	50000	1990	42.98	9.77	Refugee camp
Africa	Ethiopia	Addis Ababa	Dimnia	10756	1990	35.28	6.35	Refugee camp
Africa	Ethiopia	Addis Ababa	Fugnido	19587	1990	34.58	7.76	Refugee camp
Africa	Ethiopia	Addis Ababa	Hartishek	135000	1990	43.25	8.98	Refugee camp
Africa	Ethiopia	Addis Ababa	Kamaboker	25000	1990	44.42	8.14	Refugee camp
Africa	Ethiopia	Addis Ababa	Dharwanaji	30000	1990	42.94	8.97	Refugee camp
Africa	Ethiopia	Addis Ababa	Rebasso	10000	1990	45.77	7.99	Refugee camp
Africa	Ethiopia	Addis Ababa	Kabri Beya	15000	1990	43.06	9.12	Refugee camp
Africa	Ethiopia	Addis Ababa	Kelafo	0	0	44.3	5.7	Reception center
Africa	Ethiopia	Addis Ababa	Mustahil	0	0	44.8	5.2	Reception center
Africa	Ethiopia	Addis Ababa	Dolo	0	0	42	4.2	Reception center
Africa	Ethiopia	Addis Ababa	Sufto	0	0	41.2	4.3	Reception center
Africa	Ethiopia	Addis Ababa	Aisha	20000	1990	42.37	10.56	Refugee camp
Africa	Gabon	Libreville		0	0	9.5	0.5	Urban refugees
Africa	Gambia	Banjul		2500	0	-16.5	13.5	Mixed with local pop
Africa	Ghana	Accra	Kikor	1000	1989	0.97	5.88	
Africa	Guinea Bissau	Bissau		28000	0	-15.19	11.86	Mixed with local pop
Africa	Kenya	Nairobi	Dagahaley	39441	1990	40.11	0.9	Refugee camp
Africa	Kenya	Nairobi	El Wak	8200	1990	40.75	2.59	Refugee camp
Africa	Kenya	Nairobi	Hagadhera	43829	1990	40.5	0.166	Refugee camp
Africa	Kenya	Nairobi	Hatimy	2935	1990	39.53	-3.82	Refugee camp
Africa	Kenya	Nairobi	Liboi	48476	1990	41	0.366	Refugee camp
Africa	Kenya	Nairobi	Kakuma	29397	1990	35.24	3.49	Refugee camp
Africa	Kenya	Nairobi	Ifo	44841	1990	40.16	0.266	Refugee camp
Africa	Kenya	Nairobi	Mandera	39900	1990	41.3	3.62	Refugee camp
Africa	Kenya	Nairobi	Marafa	29392	1990	39.93	-2.65	Refugee camp
Africa	Kenya	Nairobi	Mombasa	9065	1990	39.31	-4.48	Urban refugee
Africa	Kenya	Nairobi	Thika	4700	1990	36.32	-0.99	Refugee camp
Africa	Kenya	Nairobi	Utange	42361	1990	39.35	-4.32	Refugee camp
Africa	Kenya	Nairobi	St Annes	0	0	39.4	-4.3	Reception center
Africa	Kenya	Nairobi	Banissa	18000	1990	40.67	3.92	Refugee camp
Africa	Lesotho	Maseru		0	0	28.2	-29.4	Urban refugees
Africa	Malawi	Lilongwe		950000	0	33.3	-13.9	Dispersed in all the
Africa	Mali	Bamako	Kayes	13000	0	-11.26	14.27	
Africa	Mauritania	Nouakchott	Bassikounu	15000	1991	-6.35	16.01	Refugee settlement
Africa	Mauritania	Nouakchott	Fassala	15000	1991	-5.68	15.71	Refugee settlement
Africa	Mauritania	Nouakchott	Aghor	15000	1991	-5.99	16.4	Refugee settlement
Africa	Mozambic	Maputo		0	0	32.6	-25.9	Dispersed
Africa	Namibia	Windhoek	Osiri	1000	1992	17.19	-20.59	Refugee camp
Africa	Niger	Niamey	Nguigmi	2500	0	13.07	14.15	
Africa	Rwanda	Kigali	Bugarama	7406	0	28.99	-2.7	Refugee camps
Africa	Rwanda	Kigali	Rweyeye	1496	0	29.2	-2.58	Refugee camps
Africa	Rwanda	Kigali	Ruhelu	10836	0	29.41	-2.76	Refugee camps
Africa	Rwanda	Kigali	Shorero	12837	0	28.53	-2.78	Refugee camps
Africa	Rwanda	Kigali	Uwimpfizi	10814	0	29.72	-2.76	Refugee camps
Africa	Rwanda	Kigali	Karama	16797	0	29.76	-2.77	Refugee camps

Continent	country	Capital	camps	refugee	year	longitude	latitude	comments
Africa	Rwanda	Kigali	Akankuba	3931	0	29.81	-2.7	Refugee camps
Africa	Rwanda	Kigali	Nyange	0	0	29.83	-2.62	Refugee camps
Africa	Rwanda	Kigali	Kanage	20752	0	29.881	-2.736	Refugee camps
Africa	Rwanda	Kigali	Saga 1	17902	0	29.91	-2.67	Refugee camps
Africa	Rwanda	Kigali	Saga 2	12683	0	29.9	-2.64	Refugee camps
Africa	Rwanda	Kigali	Gitsi	11514	0	29.9	-2.6	Refugee camps
Africa	Rwanda	Kigali	Rugogwe	10822	0	29.95	-2.53	Refugee camps
Africa	Rwanda	Kigali	Kagina	16181	0	29.95	-2.45	Refugee camps
Africa	Rwanda	Kigali	Ngoma	9650	0	29.9	-2.33	Refugee camps
Africa	Rwanda	Kigali	Mututu	4075	0	29.94	-2.31	Refugee camps
Africa	Rwanda	Kigali	Nyarugenge	8650	0	30	-2.28	Refugee camps
Africa	Rwanda	Kigali	Rutonde	10945	0	30.11	-2.35	Refugee camps
Africa	Rwanda	Kigali	Buronge	38312	0	30.15	-2.4	Refugee camps
Africa	Rwanda	Kigali	Nzangwa	25018	0	30.26	-2.28	Refugee camps
Africa	Rwanda	Kigali	Gafuza	1432	0	30.4	-2.21	Refugee camps
Africa	Rwanda	Kigali	Gahara	7435	0	30.56	-2.35	Refugee camps
Africa	Rwanda	Kigali	Muhura	0	0	30.5	-1.6	Refugee camp
Africa	Rwanda	Kigali	Nyamugali	0	0	30.8	-2.3	Refugee camp
Africa	Rwanda	Kigali	Nyarushishi	0	0	29.35	-2.68	Refugee camp
Africa	Rwanda	Kigali	Gisali	700	1988	29.81	-2.03	Refugee camp
Africa	Senegal	Dakar		50000	0	-17.2	14.7	Dispersed
Africa	South Africa	Pretoria	Olifants	60000	0	31.31	-23.7	
Africa	South Africa	Pretoria	Komatipoort	90000	0	31.82	-25.73	
Africa	South Africa	Pretoria	Ndume	5000	0	32.57	-27.09	
Africa	South Africa	Pretoria	Winterweld	35000	0	28.4	-25.78	
Africa	South Africa	Pretoria	Pafuri	50000	0	31.03	-22.54	
Africa	Sudan	Khartoum	Abuda	5653	1981	35.17	13.68	Area
Africa	Sudan	Khartoum	Awad el Sid	2529	1978	35.17	13.68	Area
Africa	Sudan	Khartoum	Damazne	1300	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	El Hawata	4544	1981	35.17	13.68	Area
Africa	Sudan	Khartoum	Fath el Rakham	2357	1978	35.17	13.68	Area
Africa	Sudan	Khartoum	Fau 5	3350	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Karkora New	13287	1984	35.17	13.68	Area
Africa	Sudan	Khartoum	Karkora Old	1417	1979	35.17	13.68	Area
Africa	Sudan	Khartoum	Kashmel Girba new	9001	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Kashmel Girba old	6331	1979	35.17	13.68	Area
Africa	Sudan	Khartoum	Kilo 26 New	7547	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Kilo 26 Old	4101	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Kilo 7	1450	1978	35.17	13.68	Area
Africa	Sudan	Khartoum	Kona Zeberma	33456	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Mafaza	3010	1983	35.17	13.68	Area
Africa	Sudan	Khartoum	Safawa	4223	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Shagarab 1	20196	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Shagarab 2	19909	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Shagarab 3	14955	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	South Tokar	30000	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Tawawa	16919	1980	35.17	13.68	Area
Africa	Sudan	Khartoum	Tenedba	2847	1979	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Ali	3070	1981	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Gargur	10094	1979	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Gulja	7660	1979	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Rakuba 1	4493	1984	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Rakuba 2	8200	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Rakuba 3	9672	1981	35.17	13.68	Area
Africa	Sudan	Khartoum	Um Sagatta	36000	1974	35.17	13.68	Area
Africa	Sudan	Khartoum	Wad Awad	1368	1979	35.17	13.68	Area
Africa	Sudan	Khartoum	Wad el Hileau	13606	1980	35.17	13.68	Area
Africa	Sudan	Khartoum	Wad Sherife New	59129	1985	35.17	13.68	Area
Africa	Sudan	Khartoum	Wad Sherife Old	16862	1982	35.17	13.68	Area
Africa	Sudan	Khartoum	Abu Rakham	3247	1979	35.17	13.68	Area
Africa	Swaziland	Mbabane	Ndzevane	13000	0	31.21	-26.28	Refugee settlement
Africa	Swaziland	Mbabane	Malindza	13000	0	31.33	-26.57	Reception center
Africa	Tanzania	Dar es Salaam	Kigwa	200	0	33.65	-4.75	Refugee settlement
Africa	Tanzania	Dar es Salaam	Likuyu	16700	1989	35.64	-10.35	Refugee settlement
Africa	Tanzania	Dar es Salaam	Mkuyu	800	0	37.83	-5.2	Refugee settlement

Continent	country	Capital	camps	refugee	year	longitude	latitude	comments
Africa	Tanzania	Dar es Salam	Kinuli	0	0	30.66	-1.18	Refugee settlement
Africa	Tanzania	Dar es Salam	Muyenzi	0	0	30.68	-2.58	Refugee settlement
Africa	Tanzania	Dar es Salam	Ulyankulu	0	0	32	-5	Refugee settlement
Africa	Tanzania	Dar es Salam	Mishamo	0	0	30.5	-5.5	Refugee settlement
Africa	Tanzania	Dar es Salam	Katumba	0	0	31	-5.8	Refugee settlement
Africa	Tanzania	Dar es Salam	Mwesi	0	0	30.4	-6.4	Refugee settlement
Africa	Tanzania	Dar es Salaam	Burigi	50000	1960	31.21	-2.11	Refugee settlement
Africa	Tchad	N'Djamena		0	0	15	12	Urban refugees in L
Africa	Togo	Lome	Cotonou	112000	0	1.2	6.1	Urban refugees
Africa	Uganda	Kampala	Oruchinga	0	0	30.6	-0.8	Refugee settlement
Africa	Uganda	Kampala	Nakivale	0	0	31	-0.7	Refugee settlement
Africa	Uganda	Kampala	Ibuga	0	0	29.9	0.1	Refugee settlement
Africa	Uganda	Kampala	Kahunge	0	0	30.3	0.12	Refugee settlement
Africa	Uganda	Kampala	Rwamwanja	0	0	30.5	0.3	Refugee settlement
Africa	Uganda	Kampala	Kyaka	0	0	31.2	0.5	Refugee settlement
Africa	Uganda	Kampala	Kyangwali	0	0	31	1.4	Refugee settlement
Africa	Uganda	Kampala	Kiryandongo	0	0	32.2	1.6	Refugee settlement
Africa	Uganda	Kampala	Arua	0	0	31	3	Reception center
Africa	Uganda	Kampala	Ogujebe	0	0	31.8	3	Reception center
Africa	Uganda	Kampala	Kitgum	0	0	32.8	3.3	Refugee camp
Africa	Uganda	Kampala	Mbale	0	0	34.2	1.1	Refugee camp
Africa	Uganda	Kampala	Merama	0	0	30.5	-1	Refugee settlement
Asia	Yemen	Aden	Al Kot	30000	0	46	14	Refugee camps
Africa	Zaire	Kinshasa	Doruma	14000	1990	27.8	4.8	Area
Africa	Zaire	Kinshasa	Nord Kiwu	1500	1990	29.81	-0.5	Area
Africa	Zaire	Kinshasa	Shaba	27500	1980	23.01	-10.43	Area
Africa	Zaire	Kinshasa	Sud Kiwu	50000	1991	28.37	-2.9	Area
Africa	Zaire	Kinshasa	Aba	0	0	30.3	3.9	Refugee camp
Africa	Zaire	Kinshasa	Luango	0	0	12.68	-5.61	Refugee camp
Africa	Zaire	Kinshasa	Lukula	0	0	12.89	-5.81	Refugee camp
Africa	Zaire	Kinshasa	Bas Zaire	20000	1980	14.37	-5.39	Area
Africa	Zambia	Lusaka	Mheba	25000	1990	26.21	-12.51	Refugee settlement
Africa	Zambia	Lusaka	Mayukwayukwa	6000	1990	24.42	-14.69	Refugee settlement
Africa	Zambia	Lusaka	Ukwimi	27000	1990	30.54	-14.18	Refugee settlement
Africa	Zimbabwe	Harare	Mazowe River	20000	0	32.26	-17.14	Refugee settlement
Africa	Zimbabwe	Harare	Nyngombe	20000	0	32.54	-18.35	Refugee settlement
Africa	Zimbabwe	Harare	Tongogara	20000	0	32	-20.97	Refugee settlement
Africa	Zimbabwe	Harare	Chambuta	20000	0	31.21	-21.43	Refugee settlement
Africa	Zimbabwe	Harare	Nyamitiki	20000	0	31.33	-18.47	Refugee settlement
Africa	Algeria	Algiers	Tinduff	80000	1975	-8.04	27.5	Area
Asia	Afganistan	Kabul	Sakhi Camp	23000	1993	67.06	36.42	Area
Asia	Afganistan	Kabul	Kamaz Camp	10000	1993	67.06	36.42	Area
Asia	Bengladesh	Dhaka	Balukhali 2	9915	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Dechua Palong 1	3054	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Dechua Palong 2	11287	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Dhoa Palong	12399	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Dumdania 1	14817	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Dumdania 2	2401	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Gundum 1	23504	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Gundum 2	15247	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Gundum 3	9124	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Haludia Palong	5977	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Jummapara	4679	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Kutu Palong	10530	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Maricha Palong	7646	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Nayapara 1	15108	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Nayapara 2	1843	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Nyongchari	14695	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Rangikhal	2439	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Shallerdeba	8883	1988	92.33	21.35	Area
Asia	Bengladesh	Dhaka	Balukhali 1	16911	1988	92.33	21.35	Area
Asia	Indonesia	Jakarta	Galang	12212	1986	104.4	1	Refugee camp
Asia	Malaysia	Kuala Lumpur	Sungei Besi	9524	1991	101.42	3.1	Refugee camp
Asia	Nepal	Kathmandu	Beldangi 2	18631	1990	87.42	26.66	Area
Asia	Nepal	Kathmandu	Beldangi 2 ext.	9505	1990	87.42	26.66	Area

Continent	country	Capital	camps	refugee	year	longitude	latitude	comments
Asia	Nepal	Kathmandu	Goldhap	7897	1990	87.42	28.66	Area
Asia	Nepal	Kathmandu	Khududabari 1	6981	1990	87.42	26.66	Area
Asia	Nepal	Kathmandu	Khudunabari 2	2305	1990	87.42	26.66	Area
Asia	Nepal	Kathmandu	Sanishare	16729	1990	87.42	26.66	Area
Asia	Nepal	Kathmandu	Timai	8070	1990	87.42	26.66	Area
Asia	Nepal	Kathmandu	Beldangi 1	0	1990	87.42	26.66	Area
Asia	Philippines	Manila	Puerto Princesa	4768	1992	119.4	10.5	Refugee camps
Asia	Thailand	Bangkok	Phanat Nikhom	20000	1990	101.11	13.27	Refugee camp
Asia	Thailand	Bangkok	Sikhiu	3300	1990	103.1	15.3	Refugee camp
Asia	Thailand	Bangkok	Ban Napo	11000	1990	104.7	17.3	Refugee camp
Asia	Afganistan	Kabul	Pil I Khumri	10000	1993	67.06	36.42	Area
North America	Jamaica	Kingston	Montpelier	50	1980	-77.15	18.15	
North America	Mexico	Mexico City	La Laguna	0	0	-88.2	19.5	Refugee settlement
North America	Mexico	Mexico City	Kuchumatan	0	0	-88.3	19.2	Refugee settlement
North America	Mexico	Mexico City	Los Lirios	0	0	-88.8	18.8	Refugee settlement
North America	Mexico	Mexico City	Maya Balam	0	0	-88.5	18.9	Refugee settlement
North America	Mexico	Mexico City	Quetyal Edzna	0	0	-90.2	19.8	Refugee settlement
North America	Mexico	Mexico City	Les Laureles	0	0	-90.01	19.6	Refugee settlement
North America	Mexico	Mexico City	Keste	0	0	-90.2	19.3	Refugee settlement
North America	Mexico	Mexico City	Maya Tecun	0	0	-90.6	19.2	Refugee settlement
North America	Mexico	Mexico City	State of Chiapas	0	0	-91.27	16.16	Area of 126 camps
North America	Cuba	Havana	Isla de la Juvente	50	1984	-82.9	21.8	Area

APPENDIX 2

BASIC PRINCIPLES OF GPS

These tools, which call for earthly positioning techniques through the help of geostationary satellites, enable us in a simple manner (GPS weighs only about 500 grams) to geographically locate any point on the surface of the earth. GPS gives the user a geographic position in latitude/longitude and altitude (X, Y, Z) with a precision to within 50 meters, indeed less than a meter if we are on the GPS differential system. These tools have been widely used for many years in the public domain: i.e. by navigators.

The basic theory is explained in the following pages (ref: Differential GPS explained, Trimble 1993).

Basic GPS review

2

Something this good shouldn't be free!

Before we dive into Differential GPS let's take a moment to review the basic principles behind GPS itself.

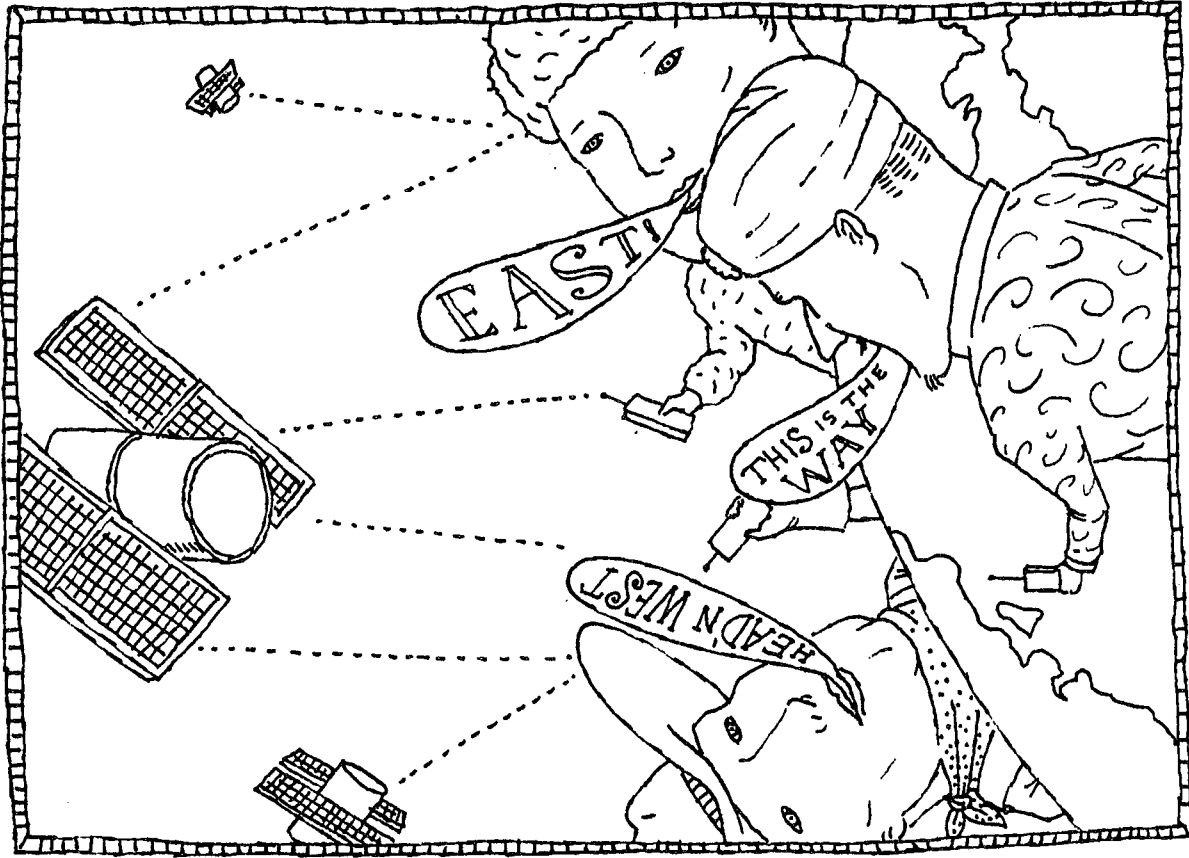
If you're just learning about navigation, or if you've been stranded on a desert island for the last few years, (perhaps because you didn't know much about navigation) you may be surprised to learn that there's an amazing new navigation technology called the Global Positioning System or "GPS."

GPS was developed by the U.S. Department of Defense as a worldwide navigation and positioning resource for both military and civilian use.

It's based on a constellation of twenty-four satellites orbiting the earth over twenty thousand kilometers up. These satellites act as reference points from which receivers on the ground "triangulate" their position. In a sense, it's like a high-tech version of the old Boy Scout technique of taking compass bearings from nearby mountain peaks to locate a point on a map.²

The satellites can act as reference points because their orbits are very accurately monitored by ground stations here on earth. By measuring the travel time of signals transmitted from the

² Purists will argue that GPS is nothing like this old Boy Scout technique because it does not use bearings to the satellites, it uses distance measurements. Purists wouldn't agree with the word "triangulate" either because it really should be "trilaterate" or perhaps "resect" . . . but you get the idea.



Basic GPS is quickly becoming the international standard for navigation—a powerful system, available to everyone, everywhere.

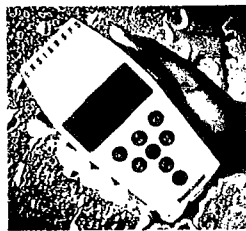
satellites, a GPS receiver on the ground can determine its distance from each satellite. And with distance measurements from four different satellites and some high-powered math, the receiver can calculate its latitude, longitude, altitude, course, and speed. In fact, good receivers can figure their position anywhere on earth to better than a hundred meters, and do it faster than once a second.³

What's more, advances in signal processing allow the faint satellite signals to be received with amazingly small antennas, so receivers can be made quite portable. In fact, some are small enough to be held in one hand.

But best of all, GPS signals are free to all users. There are no license or usage fees. With all these features, it's easy to see why GPS is quickly becoming the international standard for navigation, a powerful system, available to everyone, everywhere.

If you'd like a more complete explanation of GPS, call or write for our 80-page handbook called *GPS: A Guide to the Next Utility*.

Those of you on desert islands, just stick your request in a bottle and toss it out to sea. We'll get back to you.

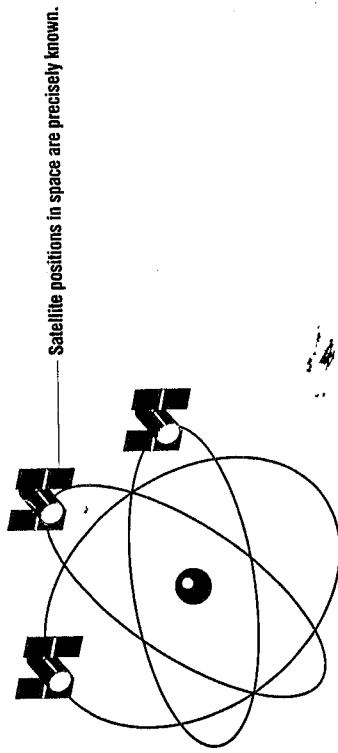


Trimble's hand-held Ensign GPS receiver.

GPS in 3 Steps

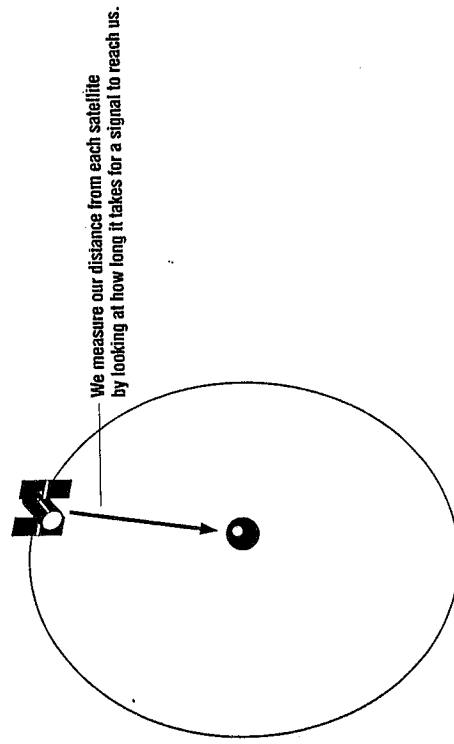
To help you get a better feel for how distance measurements to satellites can define a position, we've distilled GPS down to three basic steps:

Step 1 Satellites are reference points.



When the system is finally completed there will be 24 GPS satellites orbiting in space. Their orbital motion is constantly monitored by ground stations so their instantaneous positions are always known with great precision. Their positions are the basis of all GPS calculations.

Step 2 Signal travel time gives distance.



By listening to a series of specially coded messages transmitted by each satellite, a receiver on the ground can determine when a timing signal

3. Accuracy specs for GPS depend on many factors so no single number really tells the story. Let's use 100 meters as a conservative figure, but often GPS accuracy is much better.

for a distance of 23,000 km from the surface of the earth. The difference is the travel time for that signal. To calculate its distance to the satellite, the receiver multiplies that travel time by the speed of light:

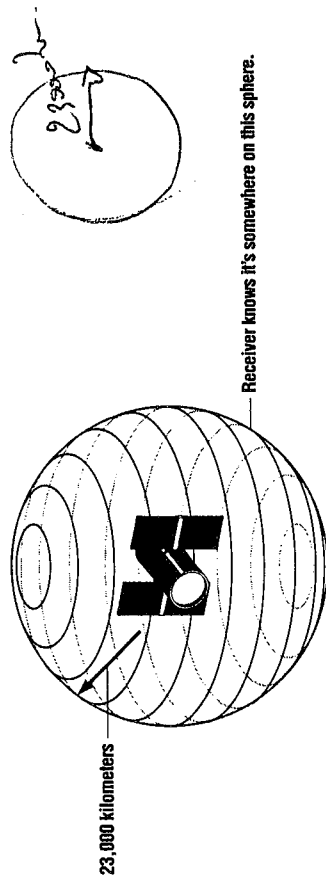
$$T_{\text{travel}} \times 3 \times 10^{10} \text{ cm/sec} = \text{Distance.}$$

This is a very important point to remember when you're trying to understand Differential GPS because it is errors in this travel time measurement that DGPS corrects.

Step 3 Three distances give position.

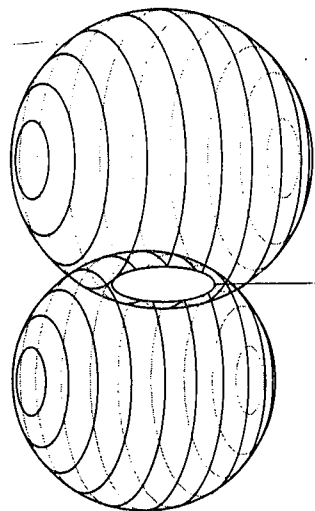
With Steps 1 and 2 accomplished, we've got distance measurements to some satellites whose positions we know exactly. Let's see how that translates into fixing our position.

Suppose a receiver determines that it is 23,000 kilometers from a particular satellite. That one measurement really narrows down where in the universe that receiver could possibly be. It tells us it is somewhere on the surface of an imaginary sphere that's centered on that satellite and that has a radius of 23,000 kilometers.



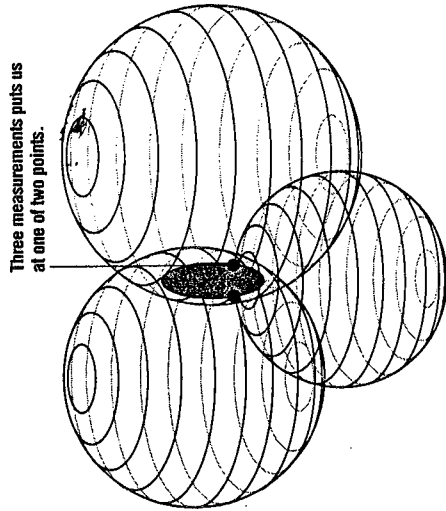
If it measures its distance to a second satellite and finds that it's 26,000 kilometers from that one, that further narrows down where it could be in space. The only places that are both 23,000 km from the first satellite and 26,000 km from the second satellite are where those two spheres intersect. That intersection is a circle of points.

A third measurement adds a third sphere which will intersect the circle formed by the other two. The intersection occurs at two points,



Two measurements put us somewhere on this circle.

and so with three measurements, the receiver has narrowed down its position to just two points in all of the universe. *Next*



Three measurements puts us at one of two points.

A fourth distance measurement would go through one of those two points but in actual practice you may not need that fourth measurement because one of the two points will be unreasonable (i.e., thousands of kilometers away from earth).

There is another reason for that fourth measurement, however. The fourth measurement gives us a way to make sure our receiver's clock is truly synchronized with universal time. But that's a long story and for that we recommend that you read our book: *GPS—A Guide to the Next Utility*.

APPENDIX 3**LISTING OF THE MAPS AVAILABLE FROM
IGN (PARIS)**

The IGN (Institut Géographique National, of France) seated in Paris, possesses, related to its cartographic mission, numerous cartographic and photographic documents on its former colonies. Part of the cartographic information is constantly updated, notably in collaboration with the national geographic institutes. This information can be accessed in Paris, at the IGN branch in Saint Mandé, after putting forth a request to the organization. Considering the historic ties that France maintains with its former colonies, the continents covered are essentially Africa (particularly West Africa), Asia (ex. Indochina: Laos, Cambodia, etc.) and the Antilles (Haiti, etc.). In the face of a lack of cartographic information on numerous developing countries, it is important to have cartographic information (even old information) on these countries. Moreover, the IGN, in collaboration with other cartographic institutes, undertook missions in numerous countries (such as the operations in Rwanda, we could recover, at PTSS, a cover at 250,000 scale in the Goma region (Zaire) and all of Rwanda, from a mission carried out by the Americans). There also exists a global cover of Rwanda in collaboration with the Belgians at 50,000 scale; it is thus a valuable source of information, and it is advisable to systematically examine it.

Contact: Mr Dupuis Chef de la Cartothèque, tel: (1) 43 98 84 12.

Other contacts from different mapping agencies all around the world :

- **Ordnance Survey** (England): Mr R. D. Fox, Tel : 44 703 792659, Fax : 44 703 792230.
- **National Defence Headquarters** (Canada): Major M. Phillips, Tel 613 996 2240, Fax 613 996 3328.
- **Defense Mapping Agency** (USA): Mr D. Dodson, Tel: 1 301 927 4759.
- **Institut Geographique National (Belgique)**: Mr Ghenot, Tel 00 32 2 648 64 80.

Figure 3a - Cartographie au 1 : 200 000 en A.O.F. et en A.E.F.. Situation fin 1960.

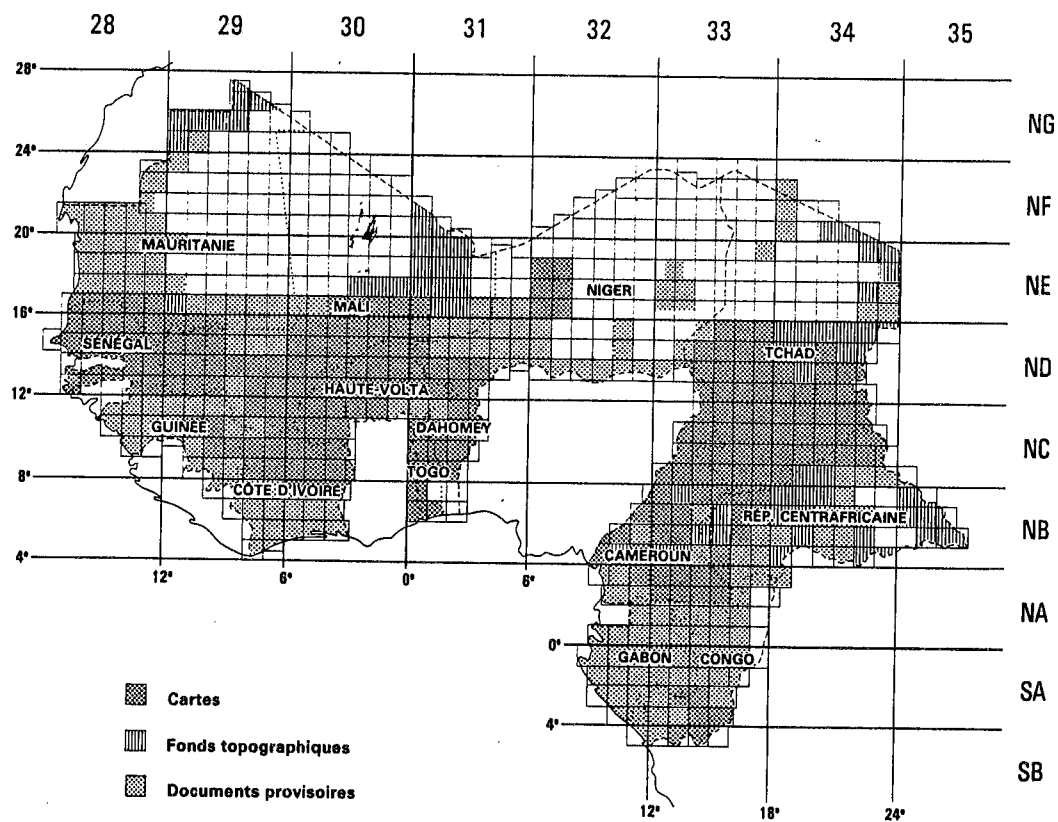


Figure 4a - Cartographie au 1 : 50 000 en A.O.F. et en A.E.F.. État d'avancement fin 1960.

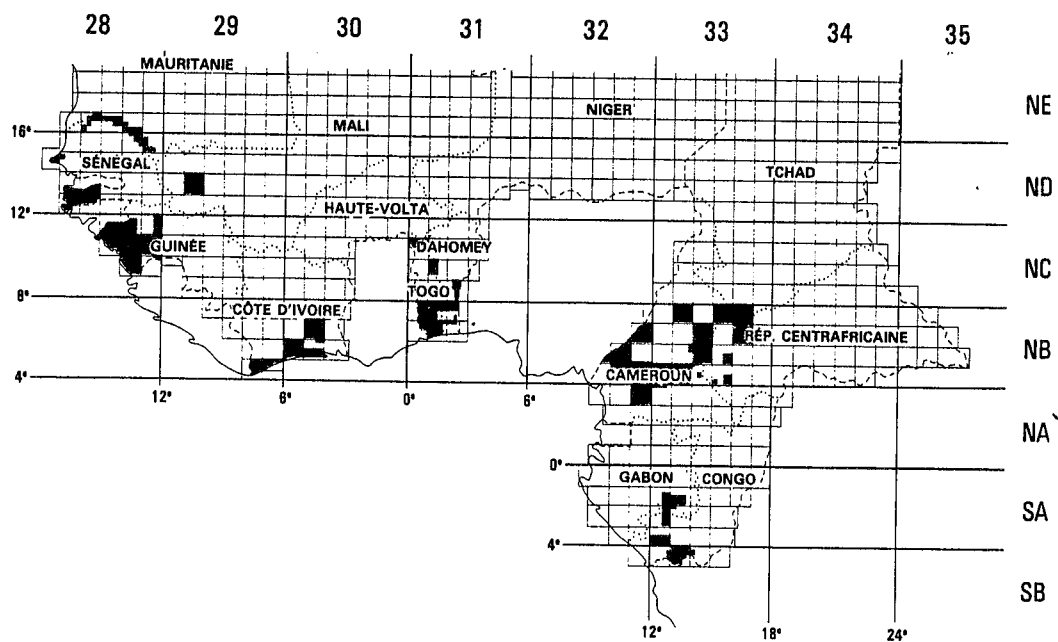


Figure 3b - Cartographie au 1 : 200 000 en Afrique de l'Ouest et en Afrique centrale.
Travaux de 1961 à fin 1985.

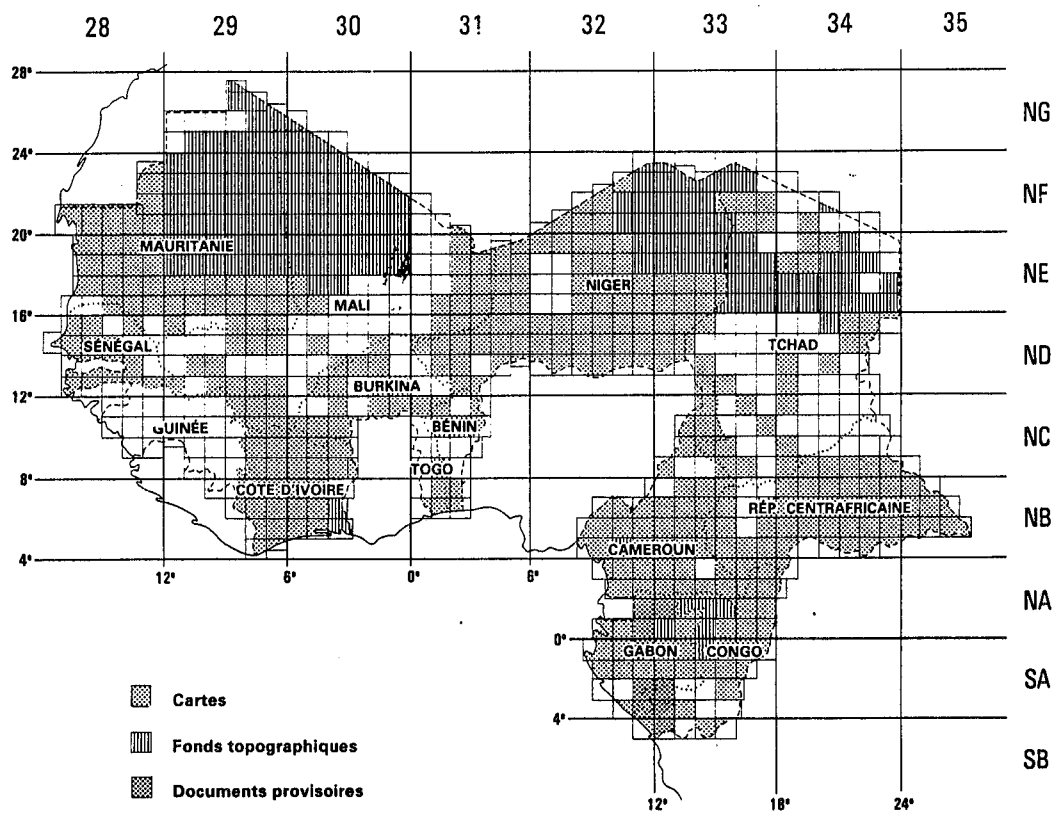
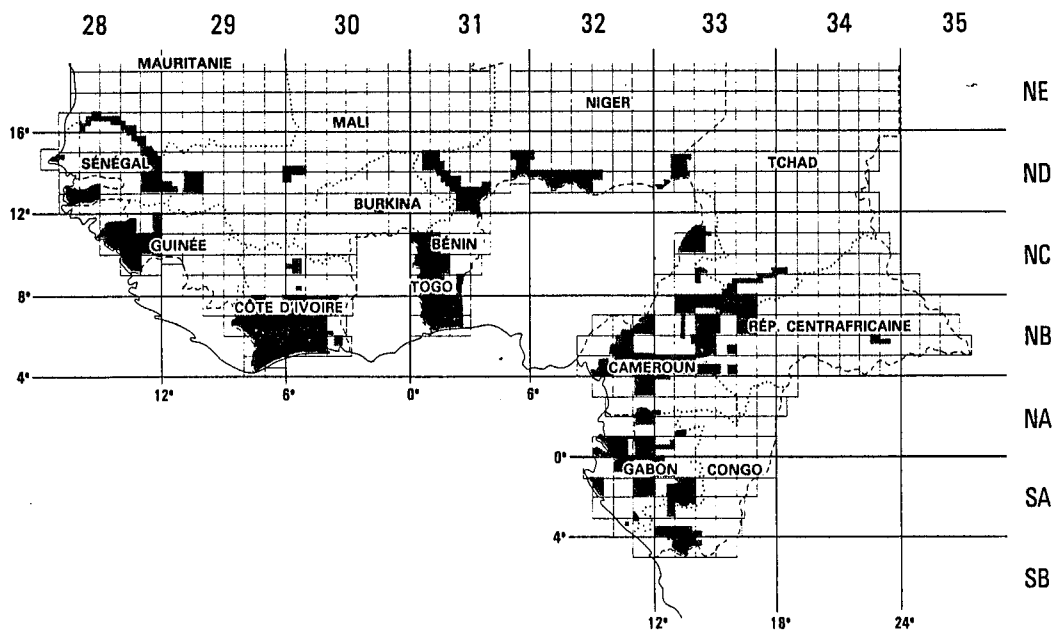
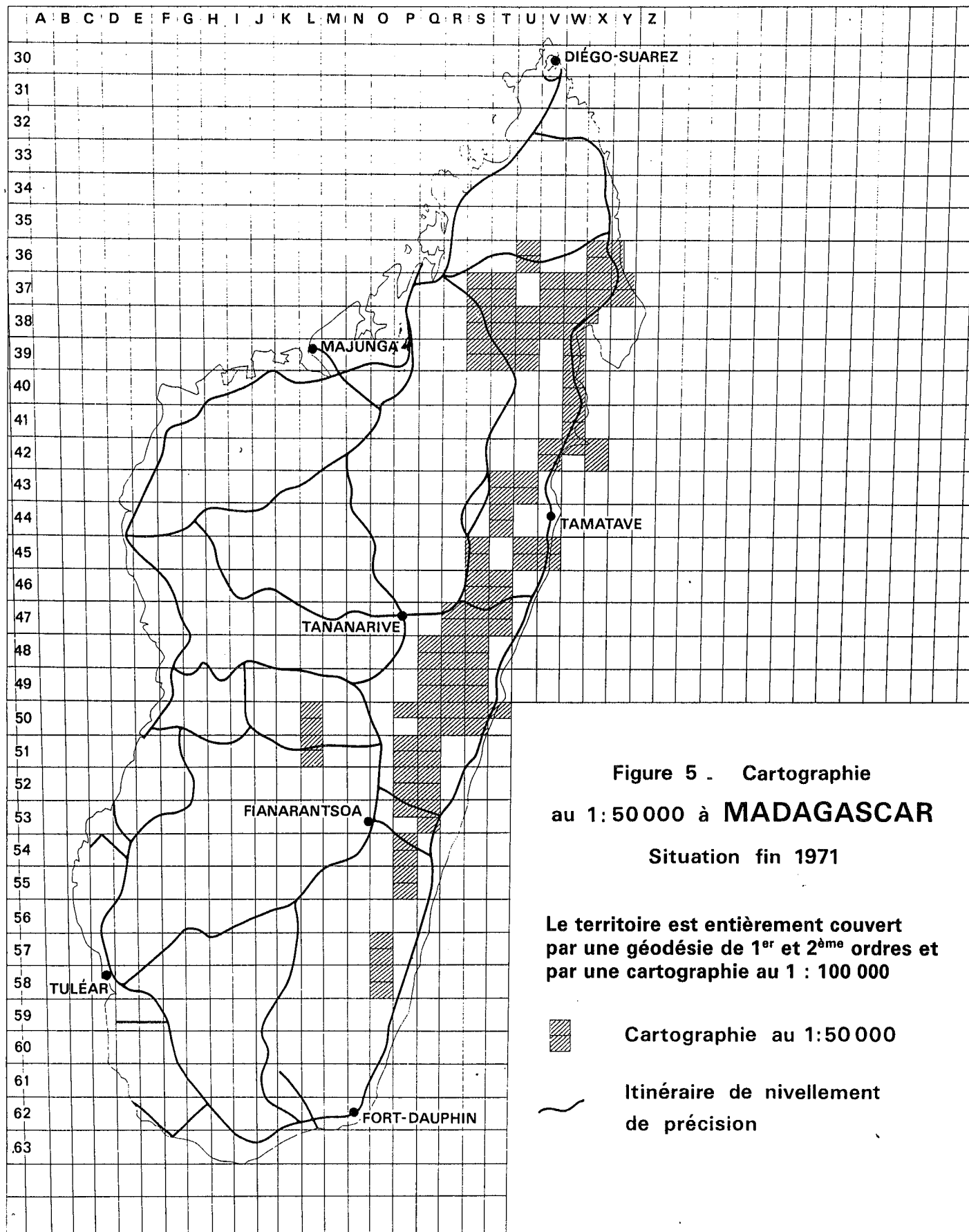
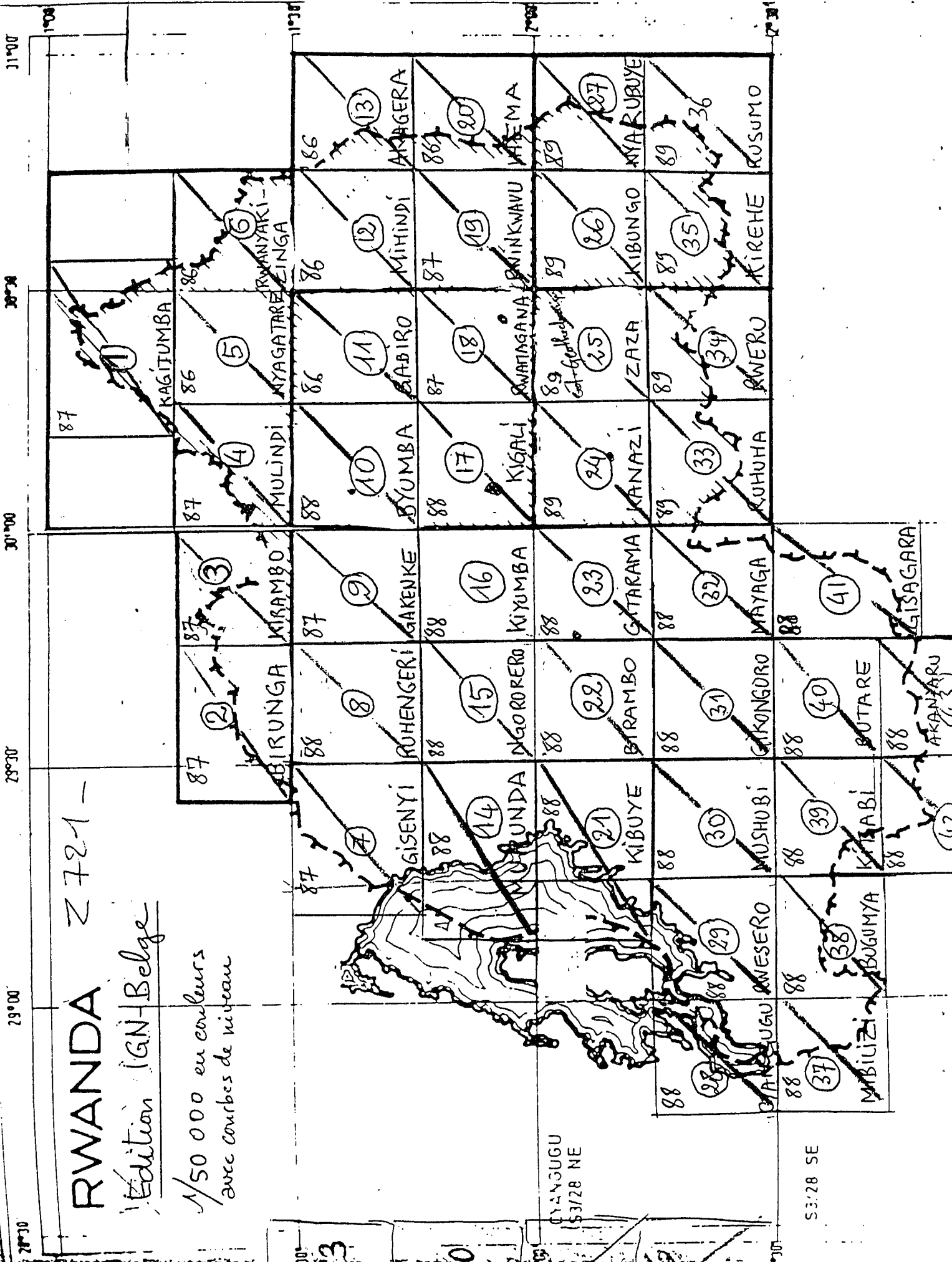


Figure 4b - Cartographie au 1 : 50 000 en Afrique de l'Ouest et en Afrique centrale.
État d'avancement fin 1985.







RWANDA Z721-

Edition IGN-Belge

1/50 000 en couleurs
avec courbes de niveau

CYANGUGU
S3/28 NE

S3/28 SE

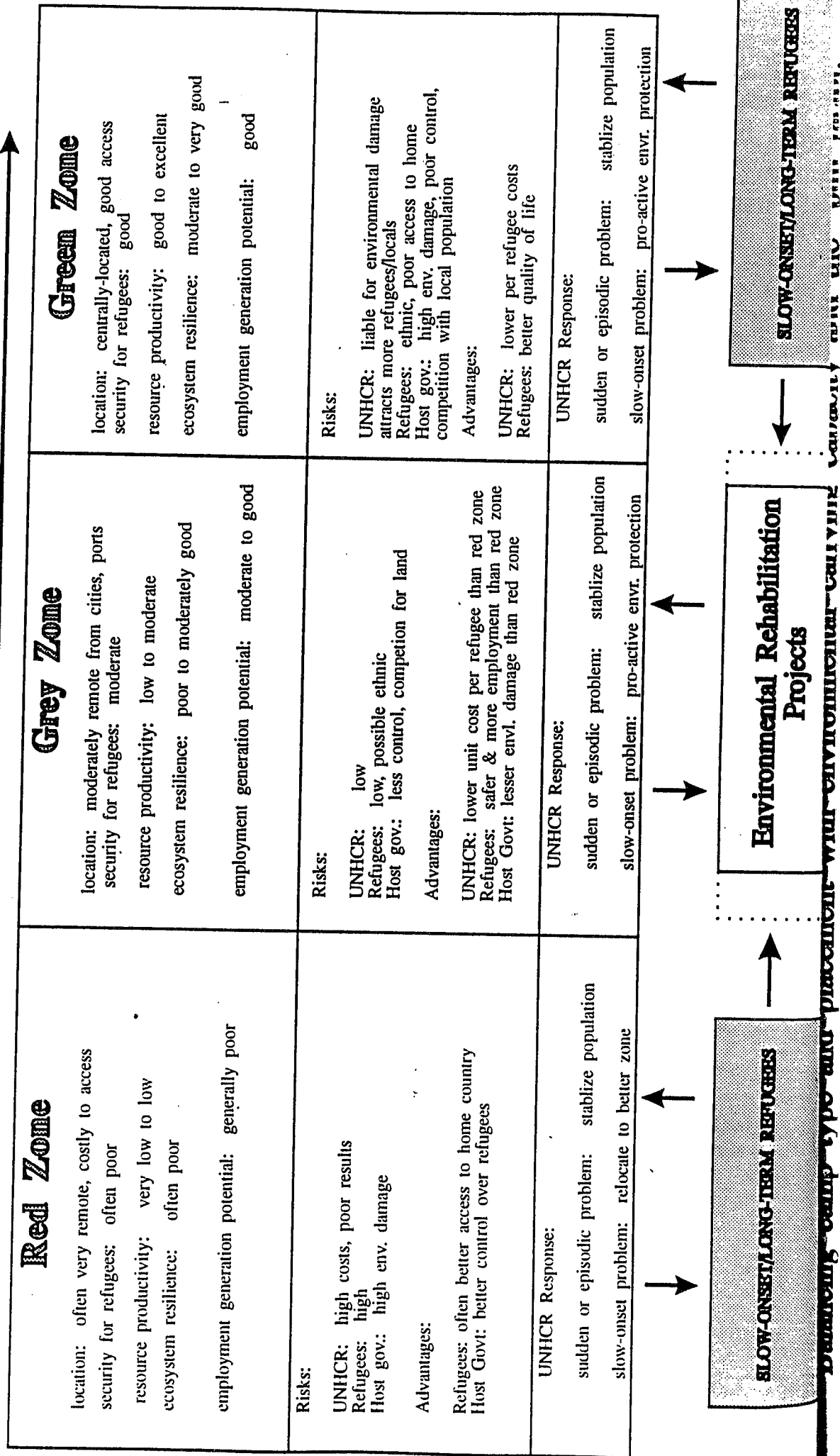
APPENDIX 4

**ENVIRONMENTAL ASSESSMENT
(J. DENNIS)**

Figure 2.1

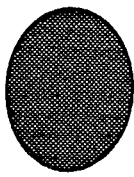
A Cost-Conscious, Environmental Carrying Capacity Approach to Siting Refugee Camps

environmental carrying capacity



Balancing camp type and placement with environmental carrying capacity and the "pull factor."

country of origin



large point of entry refugee cluster

international border

Red Zone

Red Zone

Sample strategy:

- identify labor-intensive environmental rehabilitation projects in grey zone
- negotiate resource sharing agreement with local communities/officials for example, a three-year conservation concession to establish woodlots
- convert point of entry camp into transit center to conservation camps in grey zone
- keep incentives within conservation projects low, employ food for work grant tree tenure, payable at site of final settlement

Grey Zone

Grey Zone

deforested, abandoned hillsides

river

-  transit camp
-  conservation reforestation camps
-  local community

CHAPTER TWO. A Carrying Capacity Approach to EA of Refugee Hosting Environments

UNHCR field operations are *predisposed to failure* whenever marginal people are placed on marginal sites. The simple matrix below summarizes likely outcomes:

	marginal people	empowered people
marginal site	failure	possible success
good site	possible success	success

"Success" in the above matrix is defined as saving and maintaining the lives of refugees in a manner that is cost-effective and that maintains or develops human skills and opportunity. "Success" is not just the number of people alive at the end of a fiscal year, but the number who are resettled or who are better-qualified to be successfully resettled. In a passive sense, refugees are "empowered" if their needs for water, food, shelter, and fuel are met. They are empowered in an active sense if they are given the site and/or the skills to do many or all of these things themselves.

Carrying capacity vis-a-vis refugee influxes is an ensemble of natural resources amplified by infrastructure and technology that allows a given environment to sustain a given population of persons and their livestock. Any of several variables can be limiting (shelter, energy, water, food, forage, etc.)

Environmentally-and economically-sound strategy of assistance to refugee-hosting areas (EESSARHA). Following Table 1.1 above, EA is both an extension of contingency planning and the emergency needs and resources assessment processes already established within UNHCR. Integrated with economic analysis that looks at (expected) environmental impacts and (expected) unit costs (per refugee per day) at actual and potential camp sites, EA becomes a critical component of a strategic planning process that can address the entire refugee life cycle from causes of initial displacement to the environmental training and support that will be needed to implement effective repatriation or resettlement elsewhere.

The EESSARHA MODEL.

The EESSARHA model is based on a carrying capacity approach to refugee camp management or resettlement. Rapid EA is used to estimate the capacity of various sites to support the needs of a giving refugee population as shown in the simplified site survey form below:

Camp Site or Settlement Site Name: _____
Area of associated landscape: _____ sq km
Density of local population within the associated landscape: _____ persons/sq km.
Names and populations of local communities: _____
Total local population in associated area: _____
Total present or anticipated refugee population: _____

Ability of Site and associated area to provide for local and refugee needs sustainably:

Need	local population	refugee population	total population		
Food					
Fuelwood					
Shelter					
Water -domestic -livestock -crops					

UNHCR should as a matter of policy discourage the accumulation of refugee populations densities above the estimated "carrying capacity" of a given area. Over a period of years, such accumulations in fragile, low-resilience environments can result in a downward spiral of carrying capacity and the increasing vulnerability of both refugee and local communities. Pending location-specific studies, the "default" response to an emergency refugee accumulation in a "red zone," especially in low-income food-deficit (LIFD) countries, should be to treat the initial site as a reception center for rapidly transiting refugees to "conservation camps" or other type camps in "grey zones" (see Figure 1).

The EESARHA approach to refugee camp siting would recognize both the risks of refugees remaining in harsh, fragile environments and the reluctance of host governments to allow refugees into favored areas of high productivity. **Its key innovation will be to finance the labor-intensive rehabilitation of formerly productive environments in the host country using temporary settlements of refugees (conservation corps).**

Figure 2.1 describes three zones (red, gray, and green) which assume an increasing environmental carrying capacity going from left to right. Figure 2.2 is a hypothetical map of a red and grey zone.

"Red", as in bare red soil, represents harsh environments of low innate productivity such as deserts, arid rangelands, and valleys with saline soils.

"Green", as in green plants, represents highly vegetated productive areas such as fertile, well-watered bottom lands in valley, dense forests, and rich grasslands.

"Grey", as in a zone part way between the red and green zones, represents zones that have or had moderate to good soil and water conditions, but where prolonged over-exploitation has reduced productivity. Ecologically, these areas are thought to have good "resilience" once unsustainable pressures are removed.

APPENDIX 5**LISTING OF DIGITAL DATASET AVAILABLE FROM DIFFERENT
AGENCIES, INSTITUTIONS AND PRIVATE AGENCIES**



WORLD CONSERVATION
MONITORING CENTRE



219 Huntingdon Road
Cambridge CB3 0DL, U.K.
Telephone: (0223) 277314
Telex: 817036 SCMU G
Telefax: (0223) 277136

Myriad Data Sets Available

The following list is a sample of some of the data sets available on CD-ROM of interest to the GIS community. Note that this is not a complete listing, and the brief description accompanying each entry does not attempt to provide complete information on the contents of the disc or the appropriateness of the data for a particular purpose. Please contact the vendor for complete information.

1990 Coterminous U.S. AVHRR Biweekly Composites

A series of five discs containing biweekly composite satellite images of Advanced Very High Resolution Radiometer (AVHRR) images (bands 1-5), and Normalized Difference Vegetation Index (NDVI) data. Contact: Customer Services, EROS Data Center, Sioux Falls, SD 57198, USA [605-594-6151].

Coterminous U.S. AVHRR Companion Disc

Contains several types of raster and vector data designed to work with the previous disc (but does not require it). Includes Digital Line Graph (DLG) political boundaries and water bodies, hydrologic units, climatic divisions, ecoregions, 30 arc-second Digital Elevation Model (DEM) data and an example of the 10-band Advanced Very High Resolution Radiometer biweekly composites. Contact: Customer Services, EROS Data Center, Sioux Falls, SD 57198, USA [605-594-6151].

1:2,000,000 Scale DLG Data

Digital Line Graph (DLG) coverage for the entire United States. Includes political boundaries, water bodies, railroads and major highways. Contact: Earth Science Information Center, U.S. Geological Survey, 507 National Center, Reston, VA 22092, USA [800-USA-MAPS].

Elevation, Land Cover and Population Data

U.S. Geological Survey 3 arc-second elevation data and land use/land cover data, and Bureau of the Census 1990 population database. Contact: Communication Data Services, Inc., 6105-E Arlington Blvd., Falls Church, VA 22044, USA [703-534-0034, 800-441-0034].

ArcUSA

Coverage of the coterminous United States at 1:2,000,000 scale. Data sets include political boundaries, place names, roads, railroads, water bodies, U.S. Geological Survey quad index, population and demographic attributes. All data in ARC/INFO format. Contact: Environmental Systems Research Institute, Inc., 380 New York St., Redlands, CA 92373, USA [714-793-2853, fax: 714-793-5953].

ArcWorld

Data sets include political boundaries, place name, roads, railroads, water bodies, Landsat scene index and statistical data by country. All data in ARC/INFO format. Contact: Environmental Systems Research Institute, Inc., 380 New York St., Redlands, CA 92373, USA [714-793-2853, fax: 714-793-5953].

Environmental Databases

A series of discs containing U.S. Geological Survey streamflow and water quality data, National Climatic Data Center precipitation data, and Environment Canada's surface water and sediment data. Contact: EarthInfo, Inc., 5541 Central Ave., Boulder, CO 80301, USA [303-938-1788, fax: 303-938-8183].

Digital Chart of the World

Distributed on a set of four discs, this 1,700MB data set provides global coverage at 1:1,000,000 scale in a vector format. The database includes major roads, railroads, utility networks, airports, contours, coastlines, international boundaries and populated places. Contact: *United States* — U.S. Geological Survey, ESIC-Open File Section, Box 25286, Federal Center, Mail Stop 517, Denver, CO 80225, USA [303-236-7476]. *Canada* — Products and Services Division, Surveys, Mapping and Remote Sensing Sector, Energy, Mines and Resources Canada, 615 Booth St., Room 400, Ottawa, Ontario, Canada K1A 0E4 [613-995-2123, fax: 613-995-6001]. *Europe* — Chadwyck-Healey Ltd., Cambridge Place, Cambridge CB2 1NR, United Kingdom [(0223) 311479, fax: (0223) 66440]. *Australia* — The Manager, AUSMAP Data Unit, PO Box 2, BELCONNEN ACT 2617, Australia.

Geophysics of North America

Coverage for North America, including data on magnetics, gravity, earthquake seismology, thermal aspects and stress data, vegetation index and topography. Contact: National Geophysics Data Center, NOAA E/GC, 325 Broadway, Boulder, CO 80303-3328, USA [303-497-6958, fax: 303-497-6513].

Global GRASS 1,2, & 3

A series of three discs with global coverage, including elevation, vegetation type, political boundaries, soil characteristics, marine productivity, watersheds, rivers, etc. Data are in raster format with a cell size of 4'48". Data are stored in GRASS directory structure, but the simple uncompressed file format facilitates import into other systems. Contact: Cook College Remote Sensing Center, Global Dataset Project, Box 231, College Farm Road, Rutgers University, New Brunswick, NJ 08903-0231, USA [908-932-9631, fax: 908-932-8644].

TIGER Line files

Available by state, these data include census geographic areas, streets, rivers, railroads, as well as address ranges and ZIP codes (for metro areas). Contact: Customer Services, Bureau of the Census, Washington, DC 20233, USA [301-763-4100, fax: 301-763-4794].

World CD

Global coverage of coastlines and political boundaries at 1:12,000,000 and 1:3,000,000 scales; elevation and bathymetric data at 5-minute intervals. U.S. coverage includes major roads, railroads, political boundaries and water bodies. Elevation data at 30 arc second intervals and geographic names for populated places, rivers, mountains, etc. Contact: Micro Map & CAD, 9642 W. Virginia Circle, Lakewood, CO 80226, USA [303-988-4940].

— JCC

10 Which computer systems does GRID use?
11 Which data transfer formats does GRID use?

20 What can you do to assist GRID?

GRID may be contacted via the Facility Manager at any Centre at the addresses below. To request copies of data from the GRID archive, copies of publications, to learn in more detail about GRID's activities, or to comment on *GRIDView* itself, please write, telephone, fax or e-mail:

GRID-Programme Activity Centre	Phone: + 254 2 230800 x4155
Harvey Croze	Fax: + 254 2 226491
Director	Telex: 22068 UNEP-KE
UNEP/GRID-PAC	e-mail: hcroze@nasamail.nasa.gov
Box 30552	70703,3723 Compuserve
Nairobi, KENYA	DIALCOM 41:UNE008

GRID-Tsukuba (Japan) Centre for Global Environmental Research National Institute for Environmental Studies Japan Environment Agency 16-2, Onogawa, Tsukuba Ibaraki 305, JAPAN Ph: +81 298 51611 x380 Fax: +81 298 582645	GRID-Nairobi (Kenya) UNEP Box 30552 Nairobi, KENYA Ph: + 254 2 230800 x4185 Fax: + 254 2 226491 Telex: 22068 UNEP-KE e-mail: hcroze@nasamail.nasa.gov DIALCOM 41:UNE008	ICIMOD (Nepal) Chief Administrator International Centre for Integrated Mountain Development Kathmandu, NEPAL Ph: +997 1 526313 Fax: +997 1 524509 9771	GRID-Arendal (Norway) TK-Senteret, Longum Park PO Box 1602, Myrene N-4800 Arendal, NORWAY Ph: +47 41 35500 Fax: +47 41 35050 e-mail: DIALCOM 41:UNE061 grid@grida.no 4737085050
GRID-Warsaw (Poland) ul. Jasna 2/4 00-950 Warszawa, POLAND Ph: +48 22 264231 x331 Fax: +48 22 270328 e-mail: GRIDW@PLEARN.bitnet	GRID-Geneva (Switzerland) 6, rue de la Gabelle Carouge, CH-1227 Geneva, SWITZERLAND Ph: +41 22 3438860 Fax: +41 22 3438862 Telex: 422227 GRID-CH e-mail: hebin@cgegrd11.bitnet DIALCOM 41:UNE060	GRID-Bangkok (Thailand) Outreach Building Asian Institute of Technology PO Box 2754 Bangkok 10501, THAILAND Ph: +66 2 5162124 Fax: +66 2 516 2125 e-mail: grid@ait.th DIALCOM 41:UNE096	GRID-Sioux Falls (USA) EROS Data Centre US Geological Survey Sioux Falls, SD 57198 USA Ph: +1 605 5946107 Fax: +1 605 5946589 e-mail: grid@grid1.cr.usgs.gov DIALCOM 41:UNE062

OK

May 1992

G R I D

Internet Resource List

The following Internet sites offer various software or data sets related to GIS and earth sciences:

Carbon Dioxide Information Analysis Center (CDIAC) data sets — data sets of worldwide climatic data. Access via ftp cdiac.esd.ornl.gov; cd pub.

Central Intelligence Agency (CIA) World Databank II — data file of coastlines, countries, rivers, islands and lakes of the world compiled by the CIA and now declassified into the public domain. Access via ftp hanauma.stanford.edu; cd pub/World_Map.

CIA World Factbook — copies of famous geography data sets from CIA World Databank II and CIA World Factbook. Also includes a large file of U.S. cities and their latitude/longitude coordinates. Access via ftp ftp.cs.toronto.edu; cd doc/geography.

Computer Oriented Geological Society (COGS) files — many geological programs, PC MOSS, OzGIS and pointers to data sets. Access via ftp charon.er.usgs.gov; cd pub/COGS.

Digital geologic maps — U.S. Geological Survey open-file reports of digital geologic maps, including maps of Colorado and Oregon in ARC/INFO export format. Access via ftp greenwood.cr.usgs.gov; cd pub/open-file-reports.

Earth Images — full-disk GOES satellite images are stored as Sun raster files. Also contains code that will plot the images on an orthographic projection. Access via ftp netsys.com; /pub/images.

Earthquake Information — data sets of earthquake epicenters and geologic fault traces. Access via ftp alum.wr.usgs.gov; cd /pub/map.

ETOPO5 data set — contains the ETOPO5 world elevation data set. Also Digital Elevation Models of the San Francisco area and a Macintosh mapping program. Access via ftp walrus.wr.usgs.gov; cd pub/data/dem.

FIPS 123 (Spatial Data Transfer Standard) source code — contains the current UNIX tar library of the FIPS 123 Function Library. Access via ftp resdgs1.er.usgs.gov; cd public/sdts/fipslib/unixtar

Geographic Resources Analysis Support System (GRASS) source code and data — main site for the source code to GRASS (Open GRASS Foundation, Boston, Mass.), and Spearfish data set. Connections allowed only during off-peak hours. Access via ftp moon.ccer.army.mil; cd grass.

IDRISI support software — FTP site for IDRISI (Clark University, Worcester, Mass.) support files. Access via ftp midget.towson.edu; cd /pub/gis

Map data sets — data exchange site for public-domain map data. Access via ftp spectrum.xerox.com; /pub/map.

Snow cover and flood images — National Operational Hydrologic Remote Sensing Center FTP site for satellite snow data derived from AVHRR and GOES for major portions of the United States, southern Canada and Alaska. Recently, SPOT and ERS-1 images of the Mississippi flooding appeared here. Access via ftp snow.noahrc.nws.gov; cd pub/snow.

Space Shuttle Images — Space Shuttle Earth Observation Project photography database of the Flight Science Support Office. Contains references to more than 120,000 photographs of Earth from space during the last three decades. Access via ftp sseop.jsc.nasa.gov; cd pub.

Streamflow data — contains data on U.S. Geological Survey streamflow statistics from 1874 to 1988. Access via ftp srivares.er.usgs.gov; cd hcdn92.

National Aeronautics and Space Administration Master Directory — database of high-level information about science data from Earth satellites or planetary missions, including land observations, ocean data, meteorology, astronomy, Earth/sun science, etc. Access via ftp nssdca.gsfc.nasa.gov.

National Aeronautics and Space Administration resources — planetary images from Magellan and Viking missions. News about other NASA missions. Access via ftp ames.arc.nasa.gov; cd pub/SPACE.

National Center for Atmospheric Research (NCAR) atmospheric data sets — contains information about data sets available from NCAR. Access via ftp ncardata.ucar.edu; cd pub.

National Center for Geographic Information and Analysis (NCGIA) files — NCGIA publications, initiatives, annual reports and technical reports. Annual updates to the NCGIA bibliographic database project are available in the pub/biblio directory. Digital Elevation Model data sets are in pub/dems. Access via ftp ncgia.ucsb.edu.

National Oceanic and Atmospheric Administration (NOAA)-11/12 Mosaic Images of North America — access via ftp rainbow.physics.utoronto.ca; cd pub/sat.images.

Spatial Data Transfer Standard (SDTS) documentation — all the text to the latest SDTS release in PostScript and ASCII versions. Also includes the FORTRAN source code to the General Cartographic Transformation Package (GCTP-II). Access via ftp isdres.er.usgs.gov; cd usgs/sdts or usgs/gctp.

U.S. Geological Survey mapping and coordinate transformation software — access via ftp charon.er.usgs.gov; cd pub.

Weather maps and satellite images — radar summary map GIF and PICT files, surface maps, satellite images for several U.S. cities and regions. Also hurricane images for Andrew and Emily in the subdirectory "hurricane.andrew." Access via ftp ftp.colorado.edu; pub/weather-images.

Telnet Sites

Geographic Names Server — ZIP code, population, latitude/longitude and other statistics from the 1980 census for U.S. cities. Enter name of city at the (non-existent) prompt and press Enter. Type "quit" to exit. Access via telnet martini.eecs.umich.edu 3000.

Global Land Information System — hypertext system of metadata (data about data) on various land information databases such as land use, land cover, soils, cultural and topographic data sets. Access via telnet gils.cr.usgs.gov.

National Space Science Data Center (NSSDC) on-line data — Site of the NASA Master Directory, which contains brief overview information about NASA and many important non-NASA space and earth science data, as well as data information systems. In some cases, the directory offers automatic network connections to catalogs or other systems. Access via telnet nssdca.gsfc.nasa.gov; login as NSSDC or nodis.

or information on a variety of subjects. For the GIS community, the most active list is GIS-L (see "Join the GIS-L Electronic Community," *GIS WORLD*, May 1993), where it is not uncommon to see more than 40 messages a day. However, there are also lists with titles such as *Coastal GIS*, *GIS User Interface Issues*, *GIS for German-speaking Countries*, *Geodesy*, *Maps and Air Photos*, and *Image Processing and Analysis*, as well as several groups for discussing commercial GIS software.

FTP is the service you use to download files and data sets. If your system has this service, you can log on to thousands of computers around the world that offer access and download whatever you want. There is no cost to access most FTP sites and no long-distance charges either (except perhaps an hourly connect charge assigned by your local service provider). However, files are provided "as is," and there is no guarantee that the sites always will remain open.

A rich source of map data on Internet is the site at spectrum.xerox.com, maintained by Lee Moore at Xerox, Rochester, N.Y., USA. Here you can find data sets such as U.S. Geological Survey (USGS) Digital Elevation Models (DEMs) and Digital Line Graphs (DLGs), U.S. Bureau of the Census TIGER files, USGS Geographic Names (GNIS) and Central Intelligence Agency (CIA) world data. Source code for reading the data sets also is here.

Almost all anonymous FTP sites are accessed the same way. For example, to access the spectrum site, enter ftp spectrum.xerox.com from your terminal. When the system responds with "Name," enter the word anonymous. The password is usually the word "guest" or your own e-mail address.

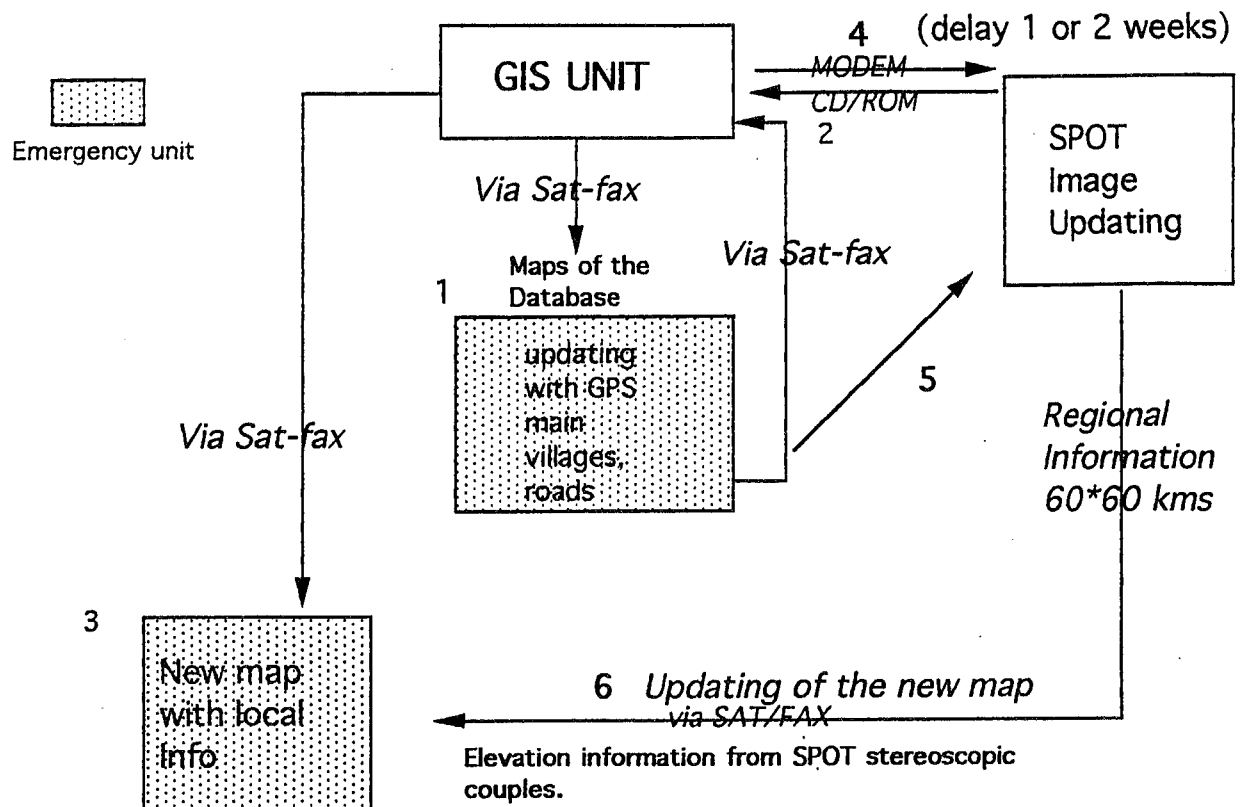
Another interesting FTP site is maintained by the National Center for Geographic Information and Analysis (NCGIA) at ncgia.ucsb.edu. At this site you can download files such as NCGIA initiatives, annual reports and the newsletter. It's also the home of the NCGIA bibliographic database of geography and GIS publications.

There are thousands of other sites at which to poke around. You can access computers that maintain stream flow data, world coastlines and national borders, weather satel-

APPENDIX 6

**PROPOSAL FOR A CARTOGRAPHY LINKED TO EMERGENCY
SITUATIONS WITHIN THE FRAMEWORK OF THE EMERGENCY
PREPAREDNESS AND RESPONSE SECTION**

EMERGENCY CARTOGRAPHY WITH GPS AND REMOTE SENSING



Total cost

- without the SPOT image : Satellite communications prices
- with the Spot images : from 1000 to 10000 Sfr (depend of the size of the image)
- Delays : from 1 days to 2 or 3 weeks with satellite images

Production of maps for Emergency Unit in UNHCR

- 1/ Extraction of the interest zone from the GIS database and fax it to the team on the field (SatFax)
- 2/ Updating of the map with GPS (via Satfax)
- 3/ Updating of the database (i.e. villages, main roads, distances, water resources...)
- 4/ Satellite imagery option (1/2 week of delay)
- 5/ Using of SPOT panchromatic as the base map for the above operations
- 6/ Using of GPS for the updating of SPOT image
- 7/ Production of Maps at the scales of 1/50000 to 1/100000°

